जब तक आपको यह परीक्षण पुस्तिका खोलने को न कहा जाए तब तक न खोलें

T.B.C.: FIAS-SFG26 **परीक्षण पुस्तिका अनुक्रम**

परीक्षण पुस्तिका

सामान्य अध्ययन

A

पूर्णांक : 100 **समय: एक घंटे**

अनुदेश

1. परीक्षा प्रारम्भ होने के तुरन्त बाद, आप इस परीक्षण पुस्तिका की पड़ताल अवश्य कर लें कि इसमें कोई बिना छपा, फटा या छूटा हुआ पृष्ठ अथवा प्रश्नांश, आदि न हो। यदि ऐसा है, तो इसे सही परीक्षण पुस्तिका से बदल लीजिए।
2. **कृपया ध्यान रखें कि OMR उत्तर-पत्रक में, उचित स्थान पर, रोल नम्बर और परिक्षण पुस्तिका अनुक्रम A, B, C या D को, ध्यान से एवं बिना किसी चूक या विसंगति के भरने और कूटबद्ध करने की जिम्मेदारी उम्मीदवार की है। किसी भी प्रकार की चूक/विसंगति की स्थिति में उत्तर -पत्रक निरस्त कर दिया जाएगा ।**
3. इस परीक्षण पुस्तिका पर साथ में दिए गए कोष्ठक में आपको अपना अनुक्रमांक लिखना है। परीक्षण पुस्तिका पर **और कुछ न** लिखें।

नाम:
ईमेल:
मोबाइल:

4. इस परीक्षण पुस्तिका में **50** प्रश्नांश (प्रश्न) हैं। प्रत्येक प्रश्नांश **अंग्रेज़ी और हिंदी** में छपा है। प्रत्येक प्रश्नांश में चार प्रत्युत्तर (उत्तर) दिए गए हैं। इनमें से एक प्रत्युतर को चुन लें, जिसे आप उत्तर-पत्रक पर अंकित करना चाहते हैं। यदि आपको ऐसा लगे कि एक से अधिक प्रत्युत्तर सही हैं, तो उस प्रत्युत्तर को अंकित करें जो आपको सर्वोत्तम लगे। किसी भी स्थिति में, प्रत्येक प्रश्नांश के लिए **केवल एक ही** प्रत्युत्तर चुनना है।
5. आपको अपने सभी प्रत्युत्तर अलग से दिए गए उत्तर-पत्रक पर ही अंकित करने हैं। उत्तर-पत्रक में दिए गए निर्देश देखिए।
6. ***सभी*** प्रश्नांशों *के अंक समान हैं।*
7. इससे पहले कि आप परीक्षण पुस्तिका के विभिन्न प्रश्नांशों के प्रत्युत्तर उत्तर-पत्रक पर अंकित करना शुरू करें, आपको प्रवेश प्रमाण-पत्र के साथ प्रेषित अनुदेशों के अनुसार कुछ विवरण उत्तर-पत्रक में देने हैं।
8. आप अपने सभी प्रत्युत्तरों को उत्तर-पत्रक में भरने के बाद तथा परीक्षा के समापन पर केवल उत्तर-पत्रक अधीक्षक को सौंप दें। आपको अपने साथ परीक्षण पुस्तिका ले जाने की अनुमति है।
9. कच्चे काम के लिए पत्रक परीक्षण पुस्तिका के अंत में संलग्न है।
10. **गलत उत्तरों के लिए दंड :**

वस्तुनिष्ठ प्रश्न-पत्रों में उम्मीदवार द्वारा दिए गए गलत उत्तरों के लिए दंड दिया जाएगा।

(i) प्रत्येक प्रश्न के लिए चार वैकल्पिक उत्तर हैं। उम्मीदवार द्वारा प्रत्येक प्रश्न के लिए दिए गए एक गलत उत्तर के लिए प्रश्न हेतु नियत किए गए अंकों का एक-तिहाई दंड के रूप में काटा जाएगा।

(ii) यदि कोई उम्मीदवार एक से अधिक उत्तर देता है, तो इसे गलत उत्तर माना जाएगा, यद्यपि दिए गए उत्तरों में से एक उत्तर सही होता है, फिर भी उस प्रश्न के लिए उपर्युक्तानुसार ही, उसी तरह का दंड दिया जाएगा।

(iii) यदि उम्मीदवार द्वारा कोई प्रश्न हल नहीं किया जाता है, अर्थात उम्मीदवार द्वारा उत्तर नहीं दिया जाता है, तो उस प्रश्न के लिए कोई दंड नहीं दिया जाएगा।

नोट: प्रश्न का उत्तर दें तथा अभ्यर्थी आत्मविश्वास स्तर (100% / 75% / 50% / 0%) का उपयोग करके उत्तर के प्रति अपनी निश्चितता का स्तर दर्ज कर सकते हैं। आप प्रत्येक प्रश्न के नीचे उपयुक्त आत्मविश्वास स्तर का चयन कर सकते हैं। इससे उन प्रश्नों की पुनरावृत्ति (रिवीजन) में सहायता मिलेगी, जिनमें उत्तर सही होने के बावजूद आप अनिश्चित थे।

जब तक आपको यह परीक्षण पुस्तिका खोलने को न कहा जाए तब तक न खोलें

Q.1) Consider the following statements:

Statement – I: Rainfall is one of the reasons for weathering of rocks.

Statement – II: Rainwater contains carbon dioxide in solution.

Statement – III: Rainwater contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

a) Both Statement-II and Statement-III are correct and both of them explain Statement-I.
b) Both Statement-II and Statement-III are correct, but only one of them explains Statement-I.
c) Only one of the Statements II and III is correct and that explains Statement-I.
d) Neither Statement -II nor Statement-III is correct

100%		75%		50%		0%	

Q.2) Consider the following phenomena:

I. The occurrence of different seasons throughout the year.
II. The bulging of the Earth at the Equator and flattening at the Poles.
III. The deflection of planetary winds to the right in the Northern Hemisphere and left in the Southern Hemisphere

How many of the above phenomena are caused by the rotation of the Earth on its own axis?

a) Only one
b) Only two
c) All the three
d) None

100%		75%		50%		0%	

Q.3) With reference to the continental drift theory proposed by Alfred Wegener, consider the following statements:

I. As per Wegener, the pole-fleeing force and tidal forces were responsible for the drifting of the continents.
II. Tillite formations found across South America, Africa, India, and Australia provide evidence in support of the theory.
III. Placer deposits in Ghana corresponding to gold-bearing veins found in Brazil provide evidence in support of the theory.

Which of the above given statements are correct?

a) I and II only
b) II and III only
c) I and III only
d) I, II and III

100%		75%		50%		0%	

Q.4) With reference to a lunar eclipse, consider the following statements:

I. It occurs when the Moon comes between the Earth and the Sun.
II. The "Ring of Fire" phenomenon is generally observed during the total lunar eclipse.
III. During a total lunar eclipse, the Moon completely appears dark.

Which of the statements given above is/are correct?

a) III only
b) II only
c) I and II only
d) None of the above

100%		75%		50%		0%	

Q.5) The term 'Laschamps Excursion', sometimes mentioned in the news in the context of Earth's history, refers to which of the following?

a) A brief global cooling event due to changes in solar energy intensity.
b) A temporary weakening and reversal of the Earth's magnetic field.
c) The periodic variation in the Earth's orbital eccentricity.
d) A massive volcanic eruption that triggered the last major Ice Age.

100%		75%		50%		0%	

Q.1) निम्नलिखित कथनों पर विचार कीजिए:

कथन-I : चट्टानों के अपक्षय के कारणों में से एक कारण वर्षा है।

कथन-II: वर्षा जल में घोल के रूप में कार्बन डाइऑक्साइड विद्यमान होता है।

कथन-III : वर्षा जल में वायुमंडलीय ऑक्सीजन विद्यमान होता है।

उपर्युक्त कथनों के संबंध में, निम्नलिखित में से कौन-सा सही है?

a) कथन-II और कथन-III दोनों सही हैं तथा दोनों, कथन-I की व्याख्या करते हैं
b) कथन-II और कथन-III दोनों सही हैं, किंतु उनमें से केवल एक, कथन-I की व्याख्या करता है।
c) कथन-II और कथन-III में से केवल एक सही है और वह कथन-I की व्याख्या करता है।
d) न तो कथन-II, न ही कथन-III सही है।

100%		75%		50%		0%	

Q.2) निम्नलिखित परिघटनाओं पर विचार कीजिए:

I. वर्ष भर विभिन्न ऋतुओं का आना।
II. भूमध्य रेखा पर पृथ्वी का उभार और ध्रुवों पर चपटा होना।
III. उत्तरी गोलार्ध में भूमंडलीय पवनों का दाईं ओर और दक्षिणी गोलार्ध में बाईं ओर विक्षेपण।

उपर्युक्त परिघटनाओं में से कितनी पृथ्वी के अपनी धुरी पर घूर्णन के कारण होती हैं?

a) केवल एक
b) केवल दो
c) सभी तीन
d) कोई नहीं

100%		75%		50%		0%	

Q.3) अल्फ्रेड वेगनर द्वारा प्रस्तावित महाद्वीपीय विस्थापन सिद्धांत के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. वेगनर के अनुसार, महाद्वीपों के विस्थापन हेतु ध्रुवीय पलायन बल (पोलर फ्लीइंग बल) और ज्वारीय बल उत्तरदायी थे।
II. दक्षिण अमेरिका, अफ्रीका, भारत और ऑस्ट्रेलिया में पाए जाने वाले टिलाइट निर्माण इस सिद्धांत के समर्थन में प्रमाण प्रदान करते हैं।
III. घाना में प्लेसर निक्षेप, जो ब्राजील में पाई जाने वाली स्वर्ण-युक्त अवनालिकाओं से मेल खाते हैं, सिद्धांत के समर्थन में प्रमाण प्रदान करते हैं।

उपर्युक्त कथनों में से कौन-से सही हैं?

a) केवल I और II
b) केवल II और III
c) केवल I और III
d) I, II और III

100%		75%		50%		0%	

Q.4) चंद्र ग्रहण के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह तब होता है जब चंद्रमा पृथ्वी और सूर्य के बीच आ जाता है।
II. पूर्ण चंद्र ग्रहण के दौरान सामान्यतः "रिंग ऑफ फायर" की घटना देखी जाती है।
III. पूर्ण चंद्र ग्रहण के दौरान, चंद्रमा पूरी तरह से काला दिखाई देता है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल III
b) केवल II
c) केवल I और II
d) उपर्युक्त में से कोई नहीं

100%		75%		50%		0%	

Q.5) 'लास्चैम्प्स एक्सकर्शन' (Laschamps Excursion) शब्द, जो कभी-कभी समाचारों में पृथ्वी के इतिहास के संदर्भ में उल्लेखित होता है, निम्नलिखित में से किससे संबंधित है?

a) सौर ऊर्जा की तीव्रता में परिवर्तन के कारण अल्पकालिक वैश्विक शीतलन घटना।
b) पृथ्वी के चुम्बकीय क्षेत्र का अस्थायी रूप से कमजोर होना तथा उसका उलट जाना।
c) पृथ्वी की कक्षा की उत्केन्द्रता में आवधिक परिवर्तन।
d) एक विशाल ज्वालामुखीय विस्फोट, जिसने अंतिम प्रमुख हिमयुग को प्रेरित किया।

100%		75%		50%		0%	

Q.6) In the context of hydrogeology, which of the following statements provides the most accurate description of an 'Aquifer'?

a) It is an area beneath the Earth that has been depleted due to groundwater extraction.
b) It is an underground layer of water-bearing rock, from which groundwater can be extracted.
c) It is a surface-level depression where rainwater collects to form a temporary wetland during the monsoon.
d) It is a rock layer within the Earth's crust that is both non-porous and impermeable.

100%		75%		50%		0%	

Q.7) With reference to the Solar System, consider the following statements:

I. Among the planets in the Solar System, only Earth and Mars have magnetic fields.
II. Mercury and Venus are the only planets without natural satellites.
III. Saturn is the only planet in the solar system with a mean density less than that of water.

Which of the statements given above is/are correct?

a) I, II, and III
b) II only
c) II and III only
d) I and III only

100%		75%		50%		0%	

Q.8) Consider the following:

I. Chalk
II. Limestone
III. Coal
IV. Sandstone

Which of the above is/are examples of organically formed sedimentary rocks?

a) I, II and III only
b) III only
c) II and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.9) Through which of the following countries does the Prime Meridian (0° longitude) pass?

I. France
II. Spain
III. Ghana
IV. Algeria
V. Portugal

Select the correct answer using the code given below:

a) I and II only
b) III and IV only
c) I, II, III and IV
d) I, III, IV and V

100%		75%		50%		0%	

Q.10) With reference to the Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025, which of the following statements is **incorrect**?

a) It replaces the Atomic Energy Act, 1962 and the Civil Liability for Nuclear Damage Act, 2010.
b) It allows up to 49% private equity in nuclear power projects.
c) It grants statutory status to the Atomic Energy Regulatory Board.
d) It provides a single statutory cap on the nuclear operators of Rs. 5000 cr irrespective of the reactor capacity.

100%		75%		50%		0%	

Q.11) In the northern hemisphere, the longest day of the year normally occurs in the:

a) First half of the month of June
b) Second half of the month of June
c) First half of the month of July
d) Second half of the month of July

100%		75%		50%		0%	

Q.6) जलभूविज्ञान के संदर्भ में, निम्नलिखित में से कौन सा कथन 'जलभृत' का सबसे सटीक वर्णन करता है?

a) यह पृथ्वी के नीचे का वह क्षेत्र है, जो भूजल दोहन के कारण समाप्त हो चुका है।
b) यह जलधारक शैलों की एक अधस्तलीय परत है, जिससे भूजल का दोहन किया जा सकता है।
c) यह सतही स्तर का एक अवसाद है, जहाँ वर्षा जल एकत्र होकर मानसून के दौरान अस्थायी आर्द्रभूमि का निर्माण करता है।
d) यह पृथ्वी की पपड़ी के भीतर स्थित ऐसी शैल परत है जो रंध्र विहीन और अपारगम्य दोनों होती है।

100%		75%		50%		0%	

Q.7) सौरमंडल के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. सौरमंडल के ग्रहों में केवल पृथ्वी और मंगल के पास ही चुम्बकीय क्षेत्र होते हैं।
II. बुध और शुक्र ऐसे एकमात्र ग्रह हैं जिनके कोई प्राकृतिक उपग्रह नहीं हैं।
III. शनि सौरमंडल का एकमात्र ग्रह है जिसका औसत घनत्व जल से कम है।

उपर्युक्त में से कौन सा/से कथन सही है/हैं?

a) I, II और III
b) केवल II
c) केवल II और III
d) केवल I और III

100%		75%		50%		0%	

Q.8) निम्नलिखित पर विचार कीजिए:

I. खड़िया (चॉक)
II. चूना पत्थर
III. कोयला
IV. बलुआ पत्थर

उपर्युक्त में से कौन-सा/से जैविक रूप से निर्मित अवसादी चट्टानों का/के उदाहरण है/हैं?

a) केवल I, II और III
b) केवल III
c) केवल II और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.9) प्रधान याम्योत्तर (0° देशांतर) निम्नलिखित में से किन देशों से होकर गुजरती है?

I. फ्रांस
II. स्पेन
III. घाना
IV. अल्जीरिया
V. पुर्तगाल

नीचे दिए गए कूट का उपयोग करके सही उत्तर चुनिए:

a) केवल I और II
b) केवल III और IV
c) I, II, III और IV
d) I, III, IV और V

100%		75%		50%		0%	

Q.10) भारत में बदलाव के लिए परमाणु ऊर्जा के सतत दोहन और विकास (SHANTI) अधिनियम, 2025 के संदर्भ में, निम्नलिखित में से कौन-सा कथन ***गलत*** है?

a) यह परमाणु ऊर्जा अधिनियम, 1962 और परमाणु क्षति के लिए नागरिक दायित्व अधिनियम, 2010 को प्रतिस्थापित करता है।
b) यह परमाणु ऊर्जा परियोजनाओं में 49% तक निजी इक्विटी की अनुमति देता है।
c) यह परमाणु ऊर्जा नियामक बोर्ड को सांविधिक दर्जा प्रदान करता है।
d) यह रिएक्टर क्षमता के निरपेक्ष परमाणु ऑपरेटरों पर 5000 करोड़ रुपये की एकल सांविधिक सीमा प्रदान करता है।

100%		75%		50%		0%	

Q.11) उत्तरी गोलार्ध में, वर्ष का सबसे लंबा दिन आम तौर पर कब होता है?

a) जून महीने का पहला पखवाड़ा
b) जून महीने का दूसरा पखवाड़ा
c) जुलाई महीने का पहला पखवाड़ा
d) जुलाई महीने का दूसरा पखवाड़ा

100%		75%		50%		0%	

Q.12) This mineral forms under extremely high pressure and temperature in Earth's mantle transition zone. Recent research has revealed that this mineral holds an enormous reservoir of water potentially larger than all surface oceans combined. The water is chemically bound within the mineral's crystal structure rather than existing as a free-flowing liquid.
Which of the following minerals is described above?
a) Perovskite
b) Ringwoodite
c) Wadsleyite
d) Pyroxenes

100%		75%		50%		0%	

Q.13) Consider the following statements:
Statement I: Felsic magma is more viscous than mafic magma.
Statement II: Felsic magma has a higher silica content than mafic magma.
Which of the following is correct in respect of the above statements?
a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is not correct.
d) Statement I is not correct but Statement II is correct.

100%		75%		50%		0%	

Q.14) With reference to igneous rocks, consider the following statements:
I. They do not typically contain fossils.
II. They are typically characterised by the lineated structure of minerals.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.15) A spacecraft launched from Earth is traveling in a straight line out into interstellar space. Which one of the following is the correct sequence of regions/boundaries it is most likely to pass through, from nearest to farthest?
I. Asteroid Belt
II. Oort Cloud
III. Heliopause
Select the correct answer using the code given below:
a) III - II - I
b) III - I - II
c) I - III - II
d) II - I - III

100%		75%		50%		0%	

Q.16) Consider the following statements:
I. The Earth's orbital speed is faster at perihelion than at aphelion.
II. The region between the Tropic of Cancer and the Tropic of Capricorn is the only zone on Earth where the Sun will ever be directly overhead.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.17) With reference to the Convection Current Theory proposed by Arthur Holmes, which of the following is the primary source of heat that drives convection cells within the Earth's mantle?
a) Residual primordial heat retained since the Earth's formation.
b) Intense pressure generated by the weight of overlying crustal rocks.
c) Radioactive decay of elements such as uranium and thorium within the mantle.
d) Frictional heat produced by tectonic plate movement along fault zones.

100%		75%		50%		0%	

Q.12) यह खनिज पृथ्वी के मेंटल संक्रमण क्षेत्र में अत्यधिक उच्च दाब एवं तापमान के अंतर्गत बनता है। हाल के शोध से यह ज्ञात हुआ है कि इस खनिज में जल का एक विशाल भंडार विद्यमान है, जो संभावित रूप से सभी सतही महासागरों के संयुक्त आकार से भी बड़ा हो सकता है। यह जल मुक्त प्रवाही द्रव के रूप में न होकर खनिज की क्रिस्टलीय संरचना में रासायनिक रूप से अवरुद्ध हुआ होता है।
निम्नलिखित में से कौन सा खनिज उपर्युक्त विवरण से संबंधित है?
a) पेरोव्स्काइट
b) रिंगवुडाइट
c) वाड्सलेयाइट
d) पाइरॉक्सीन

100%		75%		50%		0%	

Q.13) निम्नलिखित कथनों पर विचार कीजिए:
कथन I: फेल्सिक मैग्मा, मैफिक मैग्मा की तुलना में अधिक श्यान होता है।
कथन II: फेल्सिक मैग्मा में मैफिक मैग्मा की तुलना में सिलिका की मात्रा अधिक होती है।
उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन-सा सही है?
a) कथन I और कथन II दोनों सही हैं और कथन II कथन I की सही व्याख्या करता है।
b) कथन I और कथन II दोनों सही हैं लेकिन कथन II कथन I की सही व्याख्या नहीं करता है।
c) कथन I सही है लेकिन कथन II गलत है।
d) कथन I गलत है लेकिन कथन II सही है।

100%		75%		50%		0%	

Q.14) आग्नेय चट्टानों के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. उनमें आमतौर पर जीवाश्म नहीं होते हैं।
II. उनमें प्रायः खनिजों की रेखीय संरचना की विशेषता पाई जाती है।
उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?
a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.15) पृथ्वी से प्रक्षेपित एक अंतरिक्ष यान अंतरतारकीय अंतरिक्ष की ओर सीधी रेखा में अग्रसर है। निम्नलिखित में से वह किन क्षेत्रों/सीमाओं से क्रमशः (निकटतम से दूरस्थ) होकर गुजरने की सर्वाधिक संभावना रखता है?
I. क्षुद्रग्रह पेटी
II. ऊर्ट क्लाउड (Oort Cloud)
III. हेलियोपॉज़ (Heliopause)
नीचे दिए गए कूट का उपयोग करके सही उत्तर चुनिए:
a) III - II - I
b) III - I - II
c) I - III - II
d) II - I - III

100%		75%		50%		0%	

Q.16) निम्नलिखित कथनों पर विचार कीजिए:
I. पृथ्वी की कक्षीय गति अपसौर की तुलना में उपसौर पर अधिक होती है।
II. कर्क रेखा और मकर रेखा के बीच का क्षेत्र पृथ्वी पर एकमात्र ऐसा क्षेत्र है जहां सूर्य सीधे सिर के ऊपर होता है।
उपर्युक्त कथनों में से कौन सा/से सही है/हैं?
a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.17) आर्थर होम्स द्वारा प्रस्तावित संवहन धारा सिद्धांत के संदर्भ में, पृथ्वी के मेंटल के भीतर संवहन सेल को संचालित करने वाली ऊष्मा का प्राथमिक स्रोत निम्नलिखित में से कौन सा है?
a) पृथ्वी के निर्माण के समय से संचित अवशिष्ट आद्य ऊष्मा।
b) ऊपरी भूपर्पटी चट्टानों के भार से उत्पन्न तीव्र दाब।
c) मेंटल के भीतर यूरेनियम और थोरियम जैसे तत्वों का रेडियोधर्मी क्षय।
d) भ्रंश क्षेत्रों के अनुदिश विवर्तनिक प्लेटों की गति से उत्पन्न घर्षण ऊष्मा।

100%		75%		50%		0%	

Q.18) Consider the following information:

Volcano	Feature	Country
I. Mount Etna	Active Volcano	Italy
II. Mount Taftan	Extinct Volcano	Afghanistan
III. Mount Fuji	Dormant Volcano	South Korea

In how many of the above rows is the information correctly matched?
a) Only one
b) Only two
c) All the three
d) None

100%		75%		50%		0%	

Q.19) Consider the following statements with reference to the Earth's crust:
I. Continental crust is primarily composed of granite while oceanic crust is primarily composed of basalt rocks.
II. Continental crust and Oceanic crust are separated by Mohorovicic Discontinuity.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.20) With reference to the Doctrine of Clean Slate under the Insolvency and Bankruptcy framework, which of the following best describes the doctrine?
a) It allows liquidation of all assets of an insolvent company without any restructuring attempt.
b) It permits a resolved company to restart operations free from past liabilities, debts, and penalties after an approved resolution plan.
c) It grants absolute immunity to former promoters and directors from all civil and criminal proceedings.
d) It enables government agencies to reopen settled claims after resolution approval.

100%		75%		50%		0%	

Q.21) Consider the following statements:
1. The Earth's magnetic field has reversed every few hundred thousand years.
2. When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.
3. When living organisms originated, they modified the early atmosphere of the Earth.
Which of the statements given above is/are correct?
a) 1 only
b) 2 and 3 only
c) 1 and 3 only
d) 1, 2 and 3

100%		75%		50%		0%	

Q.22) Which of the following best describes the geological feature known as a 'Lacolith'?
a) A vertical or near-vertical wall-like intrusive body that cuts across the layers of the surrounding country rock.
b) A large, dome-shaped intrusive body connected by a pipe-like conduit from below.
c) A saucer-shaped intrusive body that develops a concave-upward shape due to the weight of the magma.
d) A lens-shaped mass of igneous rock that occupies the crest of an anticline or the trough of a syncline in folded strata.

100%		75%		50%		0%	

Q.23) Although the Sun is a massive celestial body, it is generally believed that it will never evolve into a Black Hole. Which of the following is the primary reason for this?
a) Sun has a high concentration of heavy metals at the core.
b) Sun lacks sufficient mass to undergo a supernova explosion.
c) The core temperature of the Sun is too high.
d) Sun has weak magnetic fields.

100%		75%		50%		0%	

Q.18) निम्नलिखित जानकारी पर विचार कीजिए:

ज्वालामुखी	विशेषता	देश
I. माउंट एटना	सक्रिय ज्वालामुखी	इटली
II. माउंट ताफ़्तान	मृत ज्वालामुखी	अफगानिस्तान
III. माउंट फ़्यूजी	निष्क्रिय ज्वालामुखी	दक्षिण कोरिया

उपर्युक्त में से कितनी पंक्तियों में दी गई जानकारी सही सुमेलित है?

a) केवल एक
b) केवल दो
c) सभी तीन
d) कोई नहीं

100%		75%		50%		0%	

Q.19) पृथ्वी की भूपर्पटी के संदर्भ में निम्नलिखित कथनों पर विचार कीजिए:

I. महाद्वीपीय भूपर्पटी मुख्य रूप से ग्रेनाइट से बनी है, जबकि महासागरीय भूपर्पटी मुख्य रूप से बेसाल्ट चट्टानों से बनी है।
II. महाद्वीपीय भूपर्पटी और महासागरीय भूपर्पटी मोहरोविसिक असम्बद्धता द्वारा विभाजित हैं।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.20) दिवाला और शोधन अक्षमता ढाँचे के अंतर्गत 'क्लीन स्लेट सिद्धांत' (Doctrine of Clean Slate) के संदर्भ में, निम्नलिखित में से कौन-सा इसका सर्वोत्तम वर्णन करता है?

a) यह किसी भी पुनर्गठन के प्रयास के बिना दिवालिया कंपनी की सभी परिसंपत्तियों के परिसमापन की अनुमति देता है।
b) यह एक समाधान प्राप्त कंपनी को स्वीकृत समाधान योजना के पश्चात पूर्व देनदारियों, ऋणों और दंडों से मुक्त होकर पुनः संचालन प्रारंभ करने की अनुमति देता है।
c) यह पूर्व प्रवर्तकों और निदेशकों को सभी दीवानी और आपराधिक कार्यवाहियों से पूर्ण प्रतिरक्षा प्रदान करता है।
d) यह सरकारी एजेंसियों को समाधान स्वीकृति के बाद निपटाए गए दावों को पुनः प्रारंभ करने में सक्षम बनाता है।

100%		75%		50%		0%	

Q.21) निम्नलिखित कथनों पर विचार कीजिए:

1. पृथ्वी का चुम्बकीय क्षेत्र हर कुछ सौ हजार सालों में उत्क्रमित हुआ है।
2. पृथ्वी जब 4000 मिलियन वर्षों से भी अधिक पहले बनी, तो ऑक्सीजन 54% थी और कार्बन डाइऑक्साइड नहीं थी ।
3. जब जीवित जीव पैदा हुए, उन्होंने पृथ्वी के आरंभिक वायुमण्डल को बदल दिया

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल 1
b) केवल 2 और 3
c) केवल 1 और 3
d) 1, 2 और 3

100%		75%		50%		0%	

Q.22) निम्नलिखित में से कौन सा विकल्प 'लैकोलिथ' नामक भूवैज्ञानिक संरचना का सबसे सटीक वर्णन करता है?

a) एक ऊर्ध्वाधर या लगभग ऊर्ध्वाधर दीवार जैसी अंतर्प्रवेशी संरचना जो आसपास की चट्टानों की परतों को काटती है।
b) एक विशाल, गुंबद के आकार की अंतर्प्रवेशी संरचना जो नीचे से पाइप जैसी नली द्वारा जुड़ी होती है।
c) एक तश्तरी के आकार की अंतर्प्रवेशी संरचना जो मैग्मा के भार के कारण ऊपर की ओर अवतल आकृति विकसित करती है।
d) आग्नेय चट्टान का एक लेंस के आकार का पिंड जो वलित चट्टानों में एंटीक्लाइन के शिखर या सिंकलाइन के गर्त में स्थित होता है।

100%		75%		50%		0%	

Q.23) हालांकि सूर्य एक विशाल खगोलीय पिंड है, फिर भी आमतौर पर यह माना जाता है कि यह कभी ब्लैक होल में परिवर्तित नहीं होगा। इसका प्राथमिक कारण निम्नलिखित में से कौन सा है?

a) सूर्य के कोर में भारी धातुओं की उच्च सांद्रता है।
b) सूर्य में सुपरनोवा विस्फोट के लिए पर्याप्त द्रव्यमान नहीं है।
c) सूर्य के कोर का तापमान बहुत अधिक है।
d) सूर्य के चुंबकीय क्षेत्र कमजोर हैं।

100%		75%		50%		0%	

Q.24) With reference to Earthquakes, consider the following statements about P-waves or the Primary Waves:

I. These are the fastest seismic waves and are the first to be recorded on a seismograph.
II. In P-waves, individual particles vibrate parallel to the direction of wave propagation.
III. They can travel only through solid material and cannot pass through liquid or gaseous material.
IV. They are the most destructive type of seismic waves and cause maximum damage on the Earth's surface.

Which of the statements given above is/are correct?

a) I, II, and IV only
b) I and II only
c) I only
d) II, III, and IV only

100%		75%		50%		0%	

Q.25) Consider the following:

1. Pyroclastic debris
2. Ash and dust
3. Nitrogen compounds
4. Sulphur compounds

How many of the above are products of volcanic eruptions?

a) Only one
b) Only two
c) Only three
d) All four

100%		75%		50%		0%	

Q.26) Consider the following statements:

Statement I: The shallow focus earthquake generally causes more damage than a deep focus earthquake.

Statement II: Compared to a deep focus earthquake, seismic waves from a shallow focus earthquake reach the Earth's surface with greater intensity due to lesser loss of energy.

Which of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is not correct.
d) Statement I is not correct but Statement II is correct.

100%		75%		50%		0%	

Q.27) Consider the following statements with respect to the process of Solifluction:

I. It is a type of mass movement that involves the gradual downslope movement of water-saturated fine grained rock debris.
II. It does not occur in permafrost regions.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.28) Large mineral grains formed in igneous rocks indicate which one of the following?

a) Rapid cooling of magma at the Earth's surface.
b) High silica content in the magma during crystallization.
c) Slow cooling of magma beneath the Earth's surface.
d) Release of volcanic gases through fissures and vents.

100%		75%		50%		0%	

Q.29) With reference to the International Date Line (IDL), consider the following statements:

I. When one crosses the International Date Line from west to east, he/she loses a day.
II. The International Date Line has no formal treaty-based legal status.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.24) भूकंप के संदर्भ में, प्राथमिक तरंगों (P-तरंगों) के बारे में निम्नलिखित कथनों पर विचार कीजिए:

I. ये सबसे तीव्र भूकंपीय तरंगें हैं और भूकंपमापी यंत्र पर सबसे पहले दर्ज की जाती हैं।

II. P-तरंगों में, कण तरंग संचरण की दिशा के समानांतर कंपन करते हैं।

III. ये केवल ठोस पदार्थों से होकर गुजर सकती हैं और तरल या गैसीय पदार्थों से होकर नहीं गुजर सकतीं।

IV. ये सबसे विनाशकारी प्रकार की भूकंपीय तरंगें हैं और पृथ्वी की सतह पर अधिकतम क्षति पहुंचाती हैं।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I, II और IV

b) केवल I और II

c) केवल I

d) केवल II, III और IV

100%		75%		50%		0%	

Q.25) निम्नलिखित पर विचार कीजिए:

1. ज्वालामुखीय मलबा
2. राख और धूल
3. नाइट्रोजन यौगिक
4. सल्फर यौगिक

उपर्युक्त में से कितने ज्वालामुखी विस्फोट के उत्पाद हैं?

a) केवल एक

b) केवल दो

c) केवल तीन

d) सभी चार

100%		75%		50%		0%	

Q.26) निम्नलिखित कथनों पर विचार कीजिए:

कथन I: उथले केंद्र वाले भूकंप आमतौर पर गहरे केंद्र वाले भूकंप की तुलना में अधिक नुकसान पहुंचाते हैं।

कथन II: गहरे केंद्र वाले भूकंप की तुलना में, उथले केंद्र वाले भूकंप से उत्पन्न भूकंपीय तरंगें कम ऊर्जा हानि के कारण पृथ्वी की सतह पर अधिक तीव्रता से पहुंचती हैं।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा कथन सही है?

a) कथन I और कथन II दोनों सही हैं और कथन II, कथन I की व्याख्या करता है।

b) कथन I और कथन II दोनों सही हैं, लेकिन कथन II, कथन I की व्याख्या नहीं करता है।

c) कथन I सही है, लेकिन कथन II सही नहीं है।

d) कथन I सही नहीं है, लेकिन कथन II सही है।

100%		75%		50%		0%	

Q.27) सोलिफ्लक्शन की प्रक्रिया के संबंध में निम्नलिखित कथनों पर विचार कीजिए:

I. यह एक प्रकार की द्रव्यमान गति है जिसमें जल से संतृप्त महीन दानेदार चट्टानी मलबे का धीरे-धीरे ढलान की ओर खिसकना शामिल है।

II. यह पर्माफ्रॉस्ट क्षेत्रों में नहीं होती है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.28) आग्नेय चट्टानों में बनने वाले बड़े खनिज कण निम्नलिखित में से किस बात का संकेत देते हैं?

a) पृथ्वी की सतह पर मैग्मा का तीव्र शीतलन।

b) क्रिस्टलीकरण के दौरान मैग्मा में सिलिका की उच्च मात्रा।

c) पृथ्वी की सतह के नीचे मैग्मा का धीमा शीतलन।

d) दरारों और रंध्रों के माध्यम से ज्वालामुखी गैसों का उत्सर्जन।

100%		75%		50%		0%	

Q.29) अंतर्राष्ट्रीय तिथि रेखा (IDL) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. जब कोई व्यक्ति पश्चिम से पूर्व की ओर अंतर्राष्ट्रीय तिथि रेखा पार करता है, तो उसे एक दिन की हानि होती है।

II. अंतर्राष्ट्रीय तिथि रेखा का कोई औपचारिक संधि-आधारित कानूनी दर्जा नहीं है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.30) The substances Mephedrone, Amphetamines, and MDMA, often seen in the news, are classified as which of the following?

a) Naturally occurring opioid substances derived from poppy latex.
b) Pesticides banned under the Rotterdam Convention.
c) Persistent Organic Pollutants (POPs) listed under the Stockholm Convention.
d) Synthetic psychotropic stimulant drugs.

100%		75%		50%		0%	

Q.31) The tail of comet is directed away from the Sun, because–

a) As the Comet rotates around the Sun, the lighter mass of the Comet is pushed away due to centrifugal force alone.
b) As the Comet rotates, the lighter mass of the Comet is attracted by some stars situated in the direction of its tail.
c) The radiation emitted by the Sun exerts a radial pressure on the comet throwing its tail away from the Sun.
d) The Tail of the Comet always exists in the same orientation.

100%		75%		50%		0%	

Q.32) Consider the following landforms/features:

I. Oceanic trenches
II. Volcanic Arc
III. Rift valleys
IV. Mid-oceanic ridges
V. Island Arc

How many of the above are primarily formed at divergent plate boundaries?

a) Only two
b) Only three
c) Only four
d) All the five

100%		75%		50%		0%	

Q.33) Consider the following discontinuities found within the Earth's interior:

I. Gutenberg Discontinuity
II. Lehmann Discontinuity
III. Conrad Discontinuity
IV. Repetti Discontinuity

Which one of the following is the correct sequence of the above discontinuities as they occur from the Earth's Crust to the Core?

a) III – IV – I – II
b) III – IV – II – I
c) IV – III – I – II
d) IV – III – II – I

100%		75%		50%		0%	

Q.34) The maximum mass that a stable white dwarf can have is known as:

a) The Roche Limit
b) The Chandrasekhar Limit
c) The Armstrong Limit
d) The Eddington Limit

100%		75%		50%		0%	

Q.35) With reference to the writ of Habeas Corpus in the Indian judiciary, consider the following statements:

I. It cannot be issued against a private individual.
II. If detention is found illegal, the court can order the immediate release of the detained person.
III. A Habeas Corpus petition cannot be entertained once a judicial remand order has been passed.
IV. It can be issued by both the Supreme Court and High Courts.

Which of the above statements given above are correct?

a) I and III only
b) II and IV only
c) II, III and IV
d) I, II and IV

100%		75%		50%		0%	

Q.30) मेफेड्रोन, एम्फेटामाइन और MDMA, जो अक्सर खबरों में दिखाई देते हैं, निम्नलिखित में से किस श्रेणी में आते हैं?

a) अफीम के लेटेक्स से प्राप्त प्राकृतिक रूप से पाए जाने वाले ओपिओइड पदार्थ।
b) रॉटरडैम कन्वेंशन के तहत प्रतिबंधित कीटनाशक।
c) स्टॉकहोम कन्वेंशन के तहत सूचीबद्ध स्थायी कार्बनिक प्रदूषक (POPs)।
d) कृत्रिम मनो-उत्तेजक दवाएं।

100%		75%		50%		0%	

Q.31) धूमकेतु की पूँछ सूर्य से दूर की ओर निर्देशित होती है, क्योंकि—

a) जैसे-जैसे धूमकेतु सूर्य के चारों ओर घूमता है, धूमकेतु का हल्का द्रव्यमान केवल अपकेंद्रीय बल के कारण दूर धकेल दिया जाता है।
b) जैसे-जैसे धूमकेतु घूमता है, धूमकेतु का हल्का द्रव्यमान उसकी पूंछ की दिशा में स्थित कुछ तारों द्वारा आकर्षित होता है।
c) सूर्य से निकलने वाला विकिरण धूमकेतु पर एक रेडियल दबाव (radial pressure) डालता है, जिससे उसकी पूंछ सूर्य से दूर चली जाती है।
d) धूमकेतु की पूंछ हमेशा एक ही दिशा में रहती है।

100%		75%		50%		0%	

Q.32) निम्नलिखित भू-आकृतियों/विशेषताओं पर विचार कीजिए:

I. महासागरीय गर्त
II. ज्वालामुखीय चाप
III. रिफ़्ट घाटियां
IV. मध्य-महासागरीय कटक
V. द्वीप चाप

उपर्युक्त में से कितनी आकृतियाँ मुख्य रूप से अपसारी प्लेट सीमाओं पर बनती हैं?

a) केवल दो
b) केवल तीन
c) केवल चार
d) सभी पाँच

100%		75%		50%		0%	

Q.33) पृथ्वी के आंतरिक भाग में पाई जाने वाली निम्नलिखित असंबद्धताओं पर विचार कीजिए:

I. गुटेनबर्ग असंबद्धता
II. लेहमैन असंबद्धता
III. कोनराड असंबद्धता
IV. रेपेटी असंबद्धता

पृथ्वी की भूपर्पटी से कोर तक इन असंबद्धताओं का सही क्रम निम्नलिखित में से कौन-सा है?

a) III — IV — I — II
b) III — IV — II — I
c) IV — III — I — II
d) IV — III — II — I

100%		75%		50%		0%	

Q.34) एक स्थिर श्वेत वामन का अधिकतम द्रव्यमान क्या कहलाता है?

a) रोश सीमा
b) चंद्रशेखर सीमा
c) आर्मस्ट्रांग सीमा
d) एडिंगटन सीमा

100%		75%		50%		0%	

Q.35) भारतीय न्यायपालिका में 'बंदी प्रत्यक्षीकरण' रिट के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. इसे किसी निजी व्यक्ति के विरुद्ध जारी नहीं किया जा सकता है।
II. यदि निरोध को अवैध पाया जाता है, तो न्यायालय निरोध में लिए गए व्यक्ति की तत्काल रिहाई का आदेश दे सकता है।
III. एक बार न्यायिक हिरासत आदेश पारित हो जाने के बाद, बंदी प्रत्यक्षीकरण याचिका पर विचार नहीं किया जा सकता है।
IV. इसे सर्वोच्च न्यायालय और उच्च न्यायालय, दोनों द्वारा जारी किया जा सकता है।

उपर्युक्त कथनों में से कौन-से सही हैं?

a) केवल I और III
b) केवल II और IV
c) II, III और IV
d) I, II और IV

100%		75%		50%		0%	

Q.36) Consider the following statements:

Statement I: In a hemisphere, the time gap between the summer solstice and the winter solstice is approximately six months.

Statement II: The Earth revolves around the Sun in an elliptical orbit, causing the distance between the Earth and the Sun to vary throughout the year.

Which one of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I
b) Both Statement I and Statement II are correct, but Statement II does not explain Statement I
c) Statement I is correct but Statement II is not correct
d) Statement I is not correct but Statement II is correct

100%		75%		50%		0%	

Q.37) Which of the following mountain ranges are formed due to oceanic-continental plate convergence?

I. Himalayas
II. Andes
III. Alps
IV. Atlas

Select the correct answer using the code given below:

a) II only
b) I and III only
c) II and IV only
d) I, II and IV

100%		75%		50%		0%	

Q.38) With reference to the Earth's Inner Core, consider the following statements:

I. It is primarily composed of iron and nickel.
II. It remains solid despite extremely high temperatures due to immense pressure at the Earth's center.
III. It rotates at a faster rate and in the opposite direction to the rest of the Earth.

Which of the statements given above are correct?

a) I and III only
b) I and II only
c) II and III only
d) I, II and III

100%		75%		50%		0%	

Q.39) Consider the following:

I. Different places on Earth experience sunrise and sunset at different times.
II. The shape of the Moon appears to be changing each night.
III. Ships disappear gradually over the horizon, with the lower part vanishing first.
IV. If poles of equal height are placed at regular intervals on level ground, their tops lie along a straight horizontal line.

How many of the above indicate that the Earth is spherical in shape?

a) Only one
b) Only two
c) Only three
d) All the four

100%		75%		50%		0%	

Q.40) Which of the following is correct about the Central Pay Commission of India?

a) It is a statutory body.
b) It operates under the Department of Expenditure, Ministry of Finance.
c) The first Pay Commission was constituted in 1950.
d) The recommendations made by the Commission are binding on the Union Government.

100%		75%		50%		0%	

Q.41) Which one of the following is a permeable rock that allows water to pass through it?

a) Granite
b) Limestone
c) Quartzite
d) Sill

100%		75%		50%		0%	

Q.36) निम्नलिखित कथनों पर विचार कीजिए:
कथन I: एक गोलार्द्ध में ग्रीष्म अयनांत और शीत अयनांत के बीच समयांतराल लगभग छह महीने होता है।
कथन II: पृथ्वी सूर्य के चारों ओर दीर्घवृत्ताकार कक्षा में परिक्रमा करती है, जिससे वर्ष भर पृथ्वी और सूर्य के बीच दूरी में परिवर्तन होता रहता है।
उपर्युक्त कथनों के संदर्भ में निम्नलिखित में से कौन-सा सही है?
a) कथन I और कथन II दोनों सही हैं तथा कथन II, कथन I की व्याख्या करता है।
b) कथन I और कथन II दोनों सही हैं, लेकिन कथन II, कथन I की व्याख्या नहीं करता है।
c) कथन I सही है, लेकिन कथन II सही नहीं है।
d) कथन I सही नहीं है, लेकिन कथन II सही है।

100%		75%		50%		0%	

Q.37) निम्नलिखित में से कौन-सी पर्वत श्रृंखलाएँ महासागरीय–महाद्वीपीय प्लेट अभिसरण के कारण बनी हैं?
I. हिमालय
II. एंडीज
III. आल्प्स
IV. एटलस
नीचे दिए गए कूट का उपयोग कर सही उत्तर का चयन कीजिए:
a) केवल II
b) केवल I और III
c) केवल II और IV
d) I, II और IV

100%		75%		50%		0%	

Q.38) पृथ्वी के आंतरिक कोर के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. यह मुख्य रूप से लोहे और निकेल से बना है।
II. पृथ्वी के केंद्र में अत्यधिक दबाव के कारण, अत्यंत उच्च तापमान होने के बावजूद यह ठोस अवस्था में रहता है।
III. यह पृथ्वी के शेष भाग की तुलना में अधिक तीव्र गति से तथा विपरीत दिशा में घूमता है।
उपर्युक्त कथनों में से कौन से सही हैं?
a) केवल I और III
b) केवल I और II
c) केवल II और III
d) I, II और III

100%		75%		50%		0%	

Q.39) निम्नलिखित पर विचार कीजिए:
I. पृथ्वी के विभिन्न स्थानों पर सूर्योदय और सूर्यास्त अलग-अलग समय पर होते हैं।
II. चंद्रमा का आकार प्रत्येक रात बदलता हुआ प्रतीत होता है।
III. जहाज़ क्षितिज के पार धीरे-धीरे गायब हो जाते हैं, जिसमें निचला भाग पहले अदृश्य होता है।
IV. यदि समान ऊँचाई के स्तम्भ समतल भूमि पर समान अंतराल पर लगाए जाएँ, तो उनके शीर्ष एक सीधी क्षैतिज रेखा में होते हैं।
उपर्युक्त में से कितने यह दर्शाते हैं कि पृथ्वी गोलाकार है?
a) केवल एक
b) केवल दो
c) केवल तीन
d) सभी चार

100%		75%		50%		0%	

Q.40) भारत के केंद्रीय वेतन आयोग के बारे में निम्नलिखित में से कौन सा कथन सही है?
a) यह एक सांविधिक निकाय है।
b) यह वित्त मंत्रालय के व्यय विभाग के अंतर्गत कार्य करता है।
c) पहले वेतन आयोग का गठन 1950 में किया गया था।
d) आयोग द्वारा की गई सिफारिशें केंद्र सरकार पर बाध्यकारी होती हैं।

100%		75%		50%		0%	

Q.41) निम्नलिखित में से कौन-सी एक ऐसी परासरणीय/पारगम्य चट्टान है जो जल को अपने माध्यम से गुजरने देती है ?
a) ग्रेनाइट
b) चूना पत्थर
c) क्वार्टजाइट
d) सिल

100%		75%		50%		0%	

Q.42) Consider the following statements:
Statement I: The duration of dawn and twilight is longer at the poles compared to the equatorial regions.
Statement II: The linear velocity of Earth's rotation is highest at the equator and decreases towards the poles.
Which one of the following is correct in respect of the above statements?
a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is not correct.
d) Statement I is not correct but Statement II is correct.

100%		75%		50%		0%	

Q.43) With reference to 'Mud volcanoes', consider the following statements:
I. These are geological formations that release mud, water, and gases instead of molten lava.
II. India's only active mud volcano is located at Baratang Island in the Andaman and Nicobar Islands.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.44) Due to the interaction between the Sun and an intruding star, a huge amount of matter was ejected from the Sun, and a cigar-shaped filament was created. This filament then started revolving around the Sun due to the effect of gravitation. With time, the filament broke into several pieces which eventually cooled and condensed to form the planets.
Which of the following theories/hypotheses is described above?
a) Steady State Theory
b) Tidal Hypothesis
c) Nebular Hypothesis
d) Binary Star Theory

100%		75%		50%		0%	

Q.45) The "Drone Wall Initiative" recently seen in the news refers to which of the following?
a) A coordinated EU–NATO defence system along Europe's Eastern flank to counter increasing drone threats, especially from Russia.
b) A maritime security initiative to deploy drones for monitoring submarine movements in the South China Sea.
c) An Indian Ministry of Defence project aimed at securing the entire length of the international border using surveillance drones.
d) An Indo-Pacific naval project deploying drone swarms for maritime surveillance.

100%		75%		50%		0%	

Q.46) If today is Monday, then which day of the week will it be exactly after $13^{11} + 14^{12} + 15^{13}$ days?
a) Monday
b) Wednesday
c) Thursday
d) Friday

100%		75%		50%		0%	

Q.47) There are three traffic signals. Each signal changes colour from green to red and then from red to green. The first signal stays green for 33 seconds, the second signal stays green for 35 seconds and the third signal stays green for 65 seconds. The duration of the red colour is exactly 15 seconds for all three signals. At 5:00 p.m., they together turn green. At what exact time will they change to red next, simultaneously?
a) 5:20:00 p.m.
b) 5:19:45 p.m.
c) 5:15:00 p.m.
d) 5:25:15 p.m.

100%		75%		50%		0%	

Q.42) निम्नलिखित कथनों पर विचार कीजिए:
कथन I: ध्रुवों पर उषाकाल (dawn) और संध्या (twilight) की अवधि भूमध्यरेखीय क्षेत्रों की तुलना में अधिक लंबी होती है।
कथन II: पृथ्वी के घूर्णन का रैखिक वेग भूमध्य रेखा पर सर्वाधिक होता है और ध्रुवों की ओर घटता जाता है।
उपर्युक्त कथनों के संदर्भ में निम्नलिखित में से कौन-सा सही है?
a) कथन I और कथन II दोनों सही हैं तथा कथन II, कथन I की व्याख्या करता है।
b) कथन I और कथन II दोनों सही हैं, लेकिन कथन II, कथन I की व्याख्या नहीं करता है।
c) कथन I सही है, लेकिन कथन II सही नहीं है।
d) कथन I सही नहीं है, लेकिन कथन II सही है।

100%		75%		50%		0%	

Q.43) 'पंकमुखी' (मड ज्वालामुखी) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. ये ऐसे भूवैज्ञानिक संरचनाएँ हैं जो द्रवित लावा के स्थान पर कीचड़, जल और गैसों का उत्सर्जन करती हैं।
II. भारत का एकमात्र सक्रिय मड ज्वालामुखी अंडमान और निकोबार द्वीपसमूह के बाराटांग द्वीप पर स्थित है।
उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?
a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.44) सूर्य और एक निकट से गुजरने वाले तारे के बीच अंतःक्रिया के कारण सूर्य से अत्यधिक मात्रा में पदार्थ बाहर निकला और एक सिगार की आकृति का तंतु (फिलामेंट) निर्मित हुआ। यह तंतु बाद में गुरुत्वाकर्षण के प्रभाव से सूर्य के चारों ओर परिक्रमा करने लगा। समय के साथ यह तंतु कई भागों में टूट गया, जो अंततः ठंडे होकर संघनित हुए और ग्रहों का निर्माण हुआ।
उपर्युक्त वर्णन निम्नलिखित में से किस सिद्धांत/परिकल्पना का है?
a) स्थिर अवस्था सिद्धांत
b) ज्वारीय परिकल्पना
c) निहारिका परिकल्पना
d) द्वैतारक सिद्धांत

100%		75%		50%		0%	

Q.45) हाल ही में समाचारों में देखा गया "ड्रोन वॉल इनिशिएटिव" निम्नलिखित में से किससे संबंधित है?
a) रूस से बढ़ते ड्रोन खतरों का मुकाबला करने के लिए यूरोप के पूर्वी मोर्चे पर एक समन्वित EU–NATO रक्षा प्रणाली।
b) दक्षिण चीन सागर में पनडुब्बी गतिविधियों की निगरानी हेतु ड्रोन तैनात करने की एक समुद्री सुरक्षा पहल।
c) निगरानी ड्रोन के माध्यम से अंतरराष्ट्रीय सीमा के संपूर्ण विस्तार को सुरक्षित करने हेतु भारत के रक्षा मंत्रालय की एक परियोजना।
d) समुद्री निगरानी के लिए ड्रोन स्वार्म तैनात करने वाली एक इंडो-पैसिफिक नौसैनिक परियोजना।

100%		75%		50%		0%	

Q.46) यदि आज सोमवार है, तो $13^{11} + 14^{12} + 15^{13}$ दिनों के ठीक बाद सप्ताह का कौन-सा दिन होगा?
a) सोमवार
b) बुधवार
c) गुरुवार
d) शुक्रवार

100%		75%		50%		0%	

Q.47) तीन ट्रैफिक सिग्नल हैं। प्रत्येक सिग्नल हरे से लाल तथा फिर लाल से हरे रंग में बदलता है। पहला सिग्नल 33 सेकंड तक हरा रहता है, दूसरा सिग्नल 35 सेकंड तक हरा रहता है और तीसरा सिग्नल 65 सेकंड तक हरा रहता है। तीनों सिग्नलों के लिए लाल रंग की अवधि ठीक 15 सेकंड है। सायं 5:00 बजे, ये सभी एक साथ हरे हो जाते हैं। वे अगली बार ठीक किस समय एक साथ लाल होंगे?
a) 5:20:00 p.m.
b) 5:19:45 p.m.
c) 5:15:00 p.m.
d) 5:25:15 p.m.

100%		75%		50%		0%	

Q.48) What is the middle term of the sequence A, Z, Z, B, B, B, Y, Y, Y, Y, C, C, C, C, C, . . ., N ?

a) I
b) J
c) Q
d) R

100%		75%		50%		0%	

Q.49) The letters of the word "SUBORDINATES" are rearranged such that the vowels are arranged alphabetically, and the consonants are arranged alphabetically, but the relative positions of the vowels and consonants remain unchanged. How many positions of the letter/letters will remain unchanged compared to the original word?

a) None
b) One
c) Two
d) Three

100%		75%		50%		0%	

Q.50) A rectangular display board measures 4.8 m in length and 2.1 m in breadth. Acoustic panels of size 130 cm by 90 cm have to be fixed such that the panels do not overlap. A panel can be placed in any orientation so long as its edges are parallel to the edges of the board. What is the maximum number of panels that can be accommodated on the board?

a) 5
b) 6
c) 7
d) 8

100%		75%		50%		0%	

Q.48) निम्नलिखित अनुक्रम का मध्य पद क्या है

A, Z, Z, B, B, B, Y, Y, Y, Y, C, C, C, C, C, . . ., N ?

a) I

b) J

c) Q

d) R

100%		75%		50%		0%	

Q.49)शब्द "SUBORDINATES" के अक्षरों को इस प्रकार पुनर्व्यवस्थित किया जाता है कि स्वर और व्यंजन वर्णानुक्रम में व्यवस्थित हो जाते हैं, लेकिन स्वर और व्यंजनों की सापेक्ष स्थिति अपरिवर्तित रहती है। मूल शब्द की तुलना में कितने अक्षरों की स्थिति अपरिवर्तित रहेगी?

a) कोई भी नहीं

b) एक

c) दो

d) तीन

100%		75%		50%		0%	

Q.50) एक आयताकार डिस्प्ले बोर्ड की लंबाई 4.8 मीटर तथा चौड़ाई 2.1 मीटर है। 130 सेमी × 90 सेमी आकार के ध्वनिक पैनलों को इस प्रकार लगाया जाना है कि वे एक-दूसरे पर अतिव्यापित न हों। पैनल को किसी भी अभिविन्यास में रखा जा सकता है, बशर्ते उसकी भुजाएं बोर्ड की भुजाओं के समानांतर हों। बोर्ड पर अधिकतम कितने पैनल लगाए जा सकते हैं?

a) 5

b) 6

c) 7

d) 8

100%		75%		50%		0%	

SPACE FOR ROUGH WORK

Q.1) Consider the following statements:
Statement – I: Rainfall is one of the reasons for weathering of rocks.
Statement – II: Rainwater contains carbon dioxide in solution.
Statement – III: Rainwater contains atmospheric oxygen.
Which one of the following is correct in respect of the above statements?
a) Both Statement-II and Statement-III are correct and both of them explain Statement-I.
b) Both Statement-II and Statement-III are correct, but only one of them explains Statement-I.
c) Only one of the Statements II and III is correct and that explains Statement-I.
d) Neither Statement -II nor Statement-III is correct

Ans) a
Exp) Option a is the correct answer.
Weathering is the action of elements of weather and climate over earth materials. There are three major groups of weathering processes: (i) chemical; (ii) physical or mechanical; (iii) biological.
Statement I is correct: The chemical weathering processes act on the rocks to decompose, dissolve or reduce them to a fine clastic state through chemical reactions by oxygen, surface and/or soil water and other acids. Water and air (oxygen and carbon dioxide) along with heat must be present to speed up all chemical reactions. **Water from rainfall is one of the reasons for weathering rocks.**
Statement II is correct: Normal rain has a pH of approximately 5.6, making it slightly acidic **due to the dissolution of carbon dioxide (CO_2), which forms weak carbonic acid. Thus, Statement II is correct and explains statement I.**
Statement III is correct: During rain, oxygen concentrations in rain drops tend to rise as rain drops saturate with oxygen as they fall. **Thus, rainwater contains atmospheric oxygen. Therefore, the rainfall supplies water, CO_2, and oxygen facilitating the chemical weathering of rocks.**
Statement II and Statement III are both correct and explain statement I.
Source: UPSC CSE Pre 2024
Subject:) Geography
Topic:) Weathering, Mass movements and Groundwater
Subtopic:)

Q.2) Consider the following phenomena:
I. The occurrence of different seasons throughout the year.
II. The bulging of the Earth at the Equator and flattening at the Poles.
III. The deflection of planetary winds to the right in the Northern Hemisphere and left in the Southern Hemisphere
How many of the above phenomena are caused by the rotation of the Earth on its own axis?
a) Only one
b) Only two
c) All the three
d) None

Ans) b
Exp) Option b is the correct answer.
Option I is incorrect: The **occurrence of seasons** is a direct result of the **Earth's revolution** around the Sun, combined with its fixed axial tilt.
Option II is correct: The **bulging at the Equator and flattening at the Poles** (the Geoid shape) is caused by **centrifugal force** generated by the Earth's rapid **rotation** on its axis.
Option III is correct: The Coriolis force, which deflects planetary winds (to the right in the Northern Hemisphere and left in the Southern Hemisphere), **is a result of the Earth's rotation.**
Source:)
https://ncert.nic.in/textbook/pdf/kegy209.pdf#:~:text=The%20force%20exerted%20by%20the,respond%20to%20the%20combined%20effect (page 78)

https://egyankosh.ac.in/bitstream/123456789/32496/1/Unit-1.pdf#:~:text=Direction%20of%20revolution%20of%20the,system%20is%20viewed%20from%20a&text=Most%20of%20the%20weather,thunderstorms%2C%20lightning%2C%20haze%2C%20fog%20etc. (page 7-8)
https://egyankosh.ac.in/bitstream/123456789/83534/3/Unit-3.pdf (page 55-56)
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.3) With reference to the continental drift theory proposed by Alfred Wegener, consider the following statements:
I. As per Wegener, the pole-fleeing force and tidal forces were responsible for the drifting of the continents.
II. Tillite formations found across South America, Africa, India, and Australia provide evidence in support of the theory.
III. Placer deposits in Ghana corresponding to gold-bearing veins found in Brazil provide evidence in support of the theory.
Which of the above given statements are correct?
a) I and II only
b) II and III only
c) I and III only
d) I, II and III

Ans) d
Exp) Option d is the correct answer.
The **Continental Drift Theory**, proposed by Alfred Wegener, postulates that **all present-day continents were once part of a single supercontinent called Pangaea, which later fragmented and drifted apart to their current positions over geological time**.
In order to support this hypothesis, Wegener gave the following evidence:

- **Statement II is correct: Tillite Evidence:** Tillite formations of similar age found across **South America, Africa, India, and Australia** indicate a **common glacial origin**, suggesting that these landmasses were once joined and located in polar regions.
- **Statement III is correct: Placer Deposit Evidence:** The occurrence of **rich placer gold deposits in Ghana corresponding to gold-bearing veins in Brazil** suggests that these regions were once contiguous, and subsequent continental separation explains the present spatial mismatch.
- **Jigsaw Fit of Continents:** The coastlines of continents, particularly the eastern coast of South America and the western coast of Africa, exhibit a **remarkable geometrical fit**, indicating that they were once joined.
- **Fossil Evidence:** Identical fossils of plants and animals (e.g., Glossopteris flora) are found across widely separated continents, suggesting **continuity of landmasses in the past**.

Statement I is correct: Wegener proposed that **continental drift was driven by forces such as the centrifugal force due to Earth's rotation (pole-fleeing force) and tidal forces caused by the gravitational pull of the Moon and the Sun**, although these explanations were later found to be inadequate.
Source:) Fundamentals of Physical Geography(Class 11th-Old NCERT) Chapter-4: Distribution of Ocean and Continents Page: 30/31
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.4) With reference to a lunar eclipse, consider the following statements:
I. It occurs when the Moon comes between the Earth and the Sun.
II. The "Ring of Fire" phenomenon is generally observed during the total lunar eclipse.
III. During a total lunar eclipse, the Moon completely appears dark.
Which of the statements given above is/are correct?
a) III only
b) II only
c) I and II only

d) None of the above

Ans) d

Exp) Option d is the correct answer.

Statement I is incorrect: In a **Lunar Eclipse**, the **Earth** comes between the Sun and the Moon. When the Moon comes between the Earth and the Sun, it is a Solar Eclipse.

A lunar eclipse occurs during the **Full Moon** phase. This is because a lunar eclipse requires the Sun, Earth, and Moon to be in a straight line (Syzygy) with the Earth in the middle, which only happens when the Moon is on the opposite side of the Earth from the Sun.

Statement II is incorrect: In an annular solar eclipse, the most notable feature is the "Ring of Fire" phenomenon.

1. During an annular solar eclipse, when the Moon passes directly in front of the Sun but is too small to cover it completely, the outer edges of the Sun remain visible as a bright ring encircling the Moon.
2. This creates a stunning visual effect often referred to as the "Ring of Fire."
3. This phenomenon is not observed in total solar eclipses, where the Moon completely covers the Sun.

Statement III is incorrect: During a total lunar eclipse, the Earth blocks direct sunlight from reaching the Moon. **One might expect the Moon to turn completely black. But it does not. Instead, it appears red or orange.** This happens because Earth's atmosphere bends and filters sunlight before it reaches the Moon.

The Earth's atmosphere bends (refracts) some sunlight. Shorter wavelengths (blue/violet) are scattered away, while longer wavelengths (**red/orange**) pass through and are reflected onto the Moon. This is the same principle of **Rayleigh Scattering** that makes sunsets look red.

Source:) https://timesofindia.indiatimes.com/science/total-lunar-eclipse-on-march-3-why-the-moon-turns-red-during-totality-and-the-science-behind-the-blood-moon/articleshow/128962142.cms
https://indianexpress.com/article/technology/science/total-lunar-eclipse-blood-moon-2026-10562645/
https://www.thehindu.com/sci-tech/science/solar-eclipse-to-create-rare-ring-of-fire-over-south-america/article68708711.ece

Subject:) Geography

Topic:) The Earth and the Universe

Subtopic:)

Q.5) The term 'Laschamps Excursion', sometimes mentioned in the news in the context of Earth's history, refers to which of the following?

a) A brief global cooling event due to changes in solar energy intensity.
b) A temporary weakening and reversal of the Earth's magnetic field.
c) The periodic variation in the Earth's orbital eccentricity.
d) A massive volcanic eruption that triggered the last major Ice Age.

Ans) b

Exp) Option b is the correct answer.

The **Laschamps Excursion** refers to a brief period in Earth's history, occurring approximately 41,000 years ago, **when the Earth's magnetic field significantly weakened and underwent a temporary reversal**. During this event, the magnetic North and South poles swapped positions for a short duration before eventually returning to their normal orientation.

Source:) https://timesofindia.indiatimes.com/life-style/health-fitness/health-news/wait-what-sunscreens-existed-41000-years-ago/articleshow/120367670.cms

Subject:) Geography

Topic:) The Earth and the Universe

Subtopic:)

Q.6) In the context of hydrogeology, which of the following statements provides the most accurate description of an 'Aquifer'?

a) It is an area beneath the Earth that has been depleted due to groundwater extraction.
b) It is an underground layer of water-bearing rock, from which groundwater can be extracted.

c) It is a surface-level depression where rainwater collects to form a temporary wetland during the monsoon.
d) It is a rock layer within the Earth's crust that is both non-porous and impermeable.

Ans) b
Exp) Option b is the correct answer.
An aquifer is a subsurface geological formation made of permeable materials (such as sand, gravel, or fractured rock) that can store and transmit groundwater, allowing extraction through wells or springs. In order to qualify as an aquifer, a rock must possess an essential degree of hydrologic properties of porosity and hydraulic conductivity. This means rock must possess necessary storativity and transmissivity to perform as an aquifer. It is distinct from impermeable layers which restrict flow, and from surface water features or volcanic structures.
Source:) https://egyankosh.ac.in/bitstream/123456789/29534/1/Unit-2.pdf Pg: 25
https://www.pib.gov.in/newsite/printrelease.aspx?relid=136334&reg=3&lang=2
Subject:) Geography
Topic:) Weathering, Mass movements and Groundwater
Subtopic:)

Q.7) With reference to the Solar System, consider the following statements:
I. Among the planets in the Solar System, only Earth and Mars have magnetic fields.
II. Mercury and Venus are the only planets without natural satellites.
III. Saturn is the only planet in the solar system with a mean density less than that of water.
Which of the statements given above is/are correct?
a) I, II, and III
b) II only
c) II and III only
d) I and III only

Ans) c
Exp) Option c is the correct answer.
Statement I is incorrect: **Most planets in the solar system have magnetic fields, though they vary in strength and origin.** Earth, Mercury, Jupiter, Saturn, Uranus, Neptune all have magnetic fields. M**ars does not have a global magnetic field,** only remnant crustal magnetism.
Statement II is correct: **Mercury and Venus are the only planets in the Solar System that do not have any natural satellites i.e. moons.** Earth has one, Mars has two (Phobos and Deimos), and the giant planets i.e. Jupiter, Saturn, Uranus, Neptune have dozens to over a hundred each.
Statement III is correct: Saturn is composed mostly of hydrogen and helium. While it is the second-largest planet, it is the least dense. **The mean density of Saturn is approximately 0.687 g/cm^3. Since the density of water is 1 g/cm^3, Saturn is light enough to float in a giant bathtub of water. It is the only planet with a density less than that of water.**
Source:) https://egyankosh.ac.in/bitstream/123456789/12213/1/Unit-11.pdf Pg:38, 42, 44
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2049766&reg=3&lang=2
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.8) Consider the following:
I. Chalk
II. Limestone
III. Coal
IV. Sandstone
Which of the above is/are examples of organically formed sedimentary rocks?
a) I, II and III only
b) III only

c) II and IV only
d) I, II, III and IV

Ans) a
Exp) Option a is the correct answer.
Options I, II and III are correct: Chalk, limestone and coal are examples of organically formed sedimentary rocks. Organically formed sedimentary rocks are formed from the remains of living organisms such as corals or shellfish, whose fleshy parts have been decomposed, leaving behind the hard shells.

- The most common rocks formed in this way are of the **calcareous type.** They include **limestones and chalk.**
- The **carbonaceous rocks** are organically **formed from vegetative matter**– swamps and forests. The pressure of overlying sediments compresses the plant remains into compact masses of carbon which eventually become **peat, lignite or coal.**

Option IV is incorrect: Sandstones are examples of mechanically formed sedimentary rocks. They are made from sand grains, often quartz fragments derived from granites. Mechanically formed sedimentary rocks have been formed from the accumulation of materials derived from other rocks which have been cemented together.
Source:) Certificate Physical and Human Geography, G.C. Leong, Chapter - 02, The Earth's Crust, Pg. 13
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust
Subtopic:)

Q.9) Through which of the following countries does the Prime Meridian (0° longitude) pass?
I. France
II. Spain
III. Ghana
IV. Algeria
V. Portugal
Select the correct answer using the code given below:
a) I and II only
b) III and IV only
c) I, II, III and IV
d) I, III, IV and V

Ans) c
Exp) Option c is the correct answer.
The Prime Meridian (0° longitude) is the imaginary line that divides the Earth into the Eastern and Western Hemispheres. It passes through three continents (Europe, Africa, and Antarctica) and eight countries. From north to south, the meridian crosses: the United Kingdom, France, Spain, Algeria, Mali, Burkina Faso, Togo, and Ghana.
Option I is correct: France: The 0° longitude line passes through western Europe and traverses France.
Option II is correct: Spain: The line passes through the eastern part of Spain.
Option III is correct: Ghana: The 0° longitude line passes through Ghana.
Option IV is correct: Algeria: Algeria is the largest country (by land area) through which the Prime Meridian passes. It enters Algeria from the Mediterranean Sea and heads south into the Sahara Desert.
Option V is incorrect: Portugal: While Portugal is at a similar latitude to Spain, it is located to the west of the Prime Meridian. The Prime Meridian lies east of Portugal and does not pass through it. Therefore, the Prime Meridian does not pass through it.
Knowledge Base:

Source:) https://egyankosh.ac.in/bitstream/123456789/80717/3/Unit-3.pdf Pg: 39, 40, 41.
https://www.thehindu.com/children/greenwich-the-historic-home-of-the-prime-meridian/article29610009.ece
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.10) With reference to the Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025, which of the following statements is **incorrect**?
a) It replaces the Atomic Energy Act, 1962 and the Civil Liability for Nuclear Damage Act, 2010.
b) It allows up to 49% private equity in nuclear power projects.
c) It grants statutory status to the Atomic Energy Regulatory Board.
d) It provides a single statutory cap on the nuclear operators of Rs. 5000 cr irrespective of the reactor capacity.

Ans) d
Exp) Option d is the correct answer.
Statement a is correct. The SHANTI Act is designed as an overarching reform statute that **replaces both the Atomic Energy Act, 1962 and the Civil Liability for Nuclear Damage Act, 2010**.
Statement b is correct. The Act allows **private sector participation with equity investment up to 49%** in nuclear power projects.
Statement c is correct. The Act grants statutory status to the Atomic Energy Regulatory Board to strengthen nuclear safety oversight.
Statement d is incorrect. In contrast to existing laws that impose a single statutory cap on operator liability, the SHANTI Bill **establishes a graded liability framework**. Under this framework, the limits on operator liability is graded from Rs. 100 cr to rs. 3000 cr, linked to reactor capacity.
Source: SFG Essential, Current Affairs, Quarterly Compilation (October–December 2025), Page: 13
Subject:) Current Affairs
Topic:) Polity
Subtopic:)

Q.11) In the northern hemisphere, the longest day of the year normally occurs in the:
a) First half of the month of June
b) Second half of the month of June
c) First half of the month of July
d) Second half of the month of July

Ans) b

Exp) Option b is the correct answer.

In the northern hemisphere, the longest day of the year is called the **summer solstice.**

The maximum amount of sunlight received by the Northern Hemisphere during this time is usually on **June 20, 21 or 22.**

The longest day of 2021 for those living north of the Equator is June 21. This day is referred to as the summer solstice, the longest day of the summer season. It occurs when the sun is directly over the Tropic of Cancer, or more specifically right over 23.5 degree north latitude.

Source: UPSC CSE Pre 2022

Subject:) Geography

Topic:) The Earth and the Universe

Subtopic:)

Q.12) This mineral forms under extremely high pressure and temperature in Earth's mantle transition zone. Recent research has revealed that this mineral holds an enormous reservoir of water potentially larger than all surface oceans combined. The water is chemically bound within the mineral's crystal structure rather than existing as a free-flowing liquid.

Which of the following minerals is described above?

a) Perovskite

b) Ringwoodite

c) Wadsleyite

d) Pyroxenes

Ans) b

Exp) Option b is the correct answer.

Option b is correct: Ringwoodite is a rare type of mineral that **forms** from olivine under very high pressures and temperatures, such as those present **in the mantle's transition zone.** Recently, researchers have unearthed an **enormous reservoir of water** that is located approximately 700 kilometers beneath the Earth's crust. This ocean is **located within a mineral called ringwoodite** and is much larger than all the surface oceans combined. The **water is chemically bound within the mineral's crystal structure rather than existing as a free-flowing liquid.**

Source:) https://timesofindia.indiatimes.com/science/scientists-discover-a-hidden-ocean-700-kilometres-beneath-earth-that-may-be-bigger-than-all-oceans/articleshow/129841690.cms

https://indianexpress.com/article/technology/science/earths-hidden-ocean/

Subject:) Geography

Topic:) Geomorphology: The Earth's Crust

Subtopic:)

Q.13) Consider the following statements:

Statement I: Felsic magma is more viscous than mafic magma.

Statement II: Felsic magma has a higher silica content than mafic magma.

Which of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I.

b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.

c) Statement I is correct but Statement II is not correct.

d) Statement I is not correct but Statement II is correct.

Ans) a

Exp) Option a is the correct answer.

Magma is not uniform in nature. Its behaviour depends on properties like composition, temperature, and volatile content. Among these, **viscosity plays a decisive role** in controlling how magma flows and how volcanic eruptions occur.

Statement I is correct. **Felsic magma** shows greater resistance to flow. It is thick and moves slowly i.e. **more viscous as compared to mafic magma,** which is relatively fluid in nature.
Statement II is correct. Higher silica content increases the **complexity of crystal arrangement** within the magma. This increases resistance to flow.
Statement II explains Statement I. As silica content increases, the **complexity in the crystal structure also increases**. This creates greater resistance within the magma. Because of this resistance, the magma does not flow easily. Thus, higher silica content directly leads to **higher viscosity**, which explains why felsic magma is more viscous than mafic magma.
Source: https://egyankosh.ac.in/bitstream/123456789/66689/1/Unit-4.pdf - Concept of Magma - Physical Properties of Magma – Page 93 and Major Types of Magmas – Page 96
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.14) With reference to igneous rocks, consider the following statements:
I. They do not typically contain fossils.
II. They are typically characterised by the lineated structure of minerals.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) a
Exp) Option a is the correct answer.
Igneous rocks originate in the initial stage of earth formation by the cooling of hot magma. Approximately, more than 80 per cent of earth's crust is made up of igneous rocks.
Statement I is correct: Igneous rocks do not contain any fossils. On the other hand, **sedimentary rocks** are formed from sediment accumulated over long periods, usually under water. They are non-crystalline and **often contain fossils** of animals, plants and other micro-organisms. Sedimentary rocks are thus the most varied in their formation of all rocks.
Statement II is incorrect: Igneous rocks do not typically exhibit a lineated structure. Lineation is a structural feature associated with metamorphic rocks in which rocks grains or minerals get arranged in layers or lines.
Source:) https://egyankosh.ac.in/bitstream/123456789/57480/1/Unit5.pdf (Pg. 12 - 14)
Certificate Physical and Human Geography, G.C. Leong, Chapter - 02, The Earth's Crust, Pg. 12- 13
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust
Subtopic:)

Q.15) A spacecraft launched from Earth is traveling in a straight line out into interstellar space. Which one of the following is the correct sequence of regions/boundaries it is most likely to pass through, from nearest to farthest?
I. Asteroid Belt
II. Oort Cloud
III. Heliopause
Select the correct answer using the code given below:
a) III - II - I
b) III - I - II
c) I - III - II
d) II - I - III

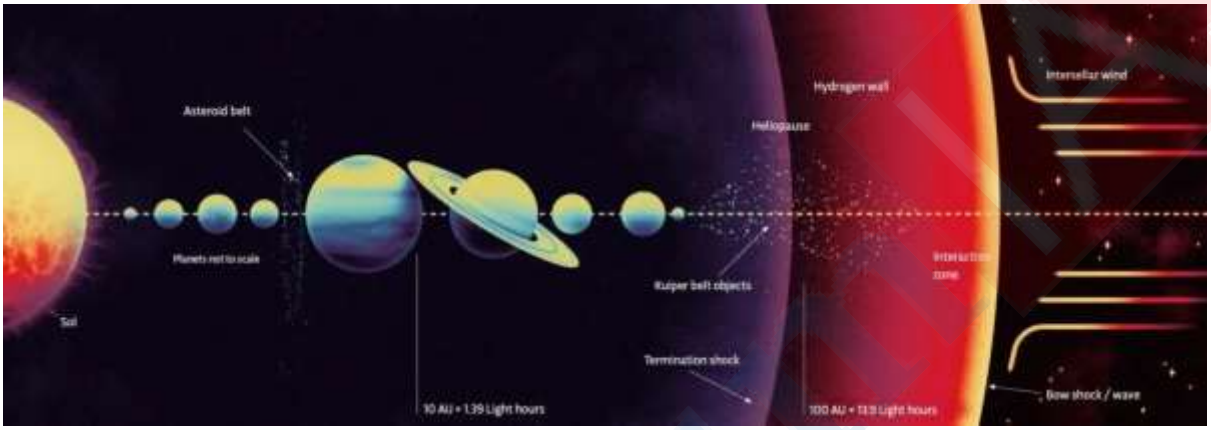
Ans) c
Exp) Option c is the correct answer.

Option I: Asteroid Belt: It is **situated between the orbits of Mars and Jupiter.** This is the first major region of debris the spacecraft encounters after leaving the inner solar system (the Terrestrial planets). It contains millions of rocky fragments and dwarf planets.

Option III: Heliopause: It is the outermost boundary of the Heliosphere. This is the "final frontier" of the Sun's influence. It is the point where the solar wind (charged particles from the Sun) is stopped by the interstellar medium.

Option II: Oort Cloud: It is a spherical shell of icy planetesimals surrounding the entire solar system. It is so distant that the Sun's light takes over a year to reach its outer edge. It is considered the source of long-period comets. **NASA's Voyager 1 and 2** passed the heliopause in 2012 and 2018 respectively, **they are expected to reach the inner edge of the Oort Cloud in roughly 300 years** and take 30,000 years to exit it.

Hence, a spacecraft moving outward from Earth will first encounter the Asteroid Belt (between Mars and Jupiter). Beyond the giant planets lies the Kuiper Belt (region of icy bodies beyond Neptune). Further outward is the Heliopause, marking the boundary where the Sun's solar wind is stopped by the interstellar medium. The outermost region is the distant Oort Cloud, a spherical shell of cometary bodies. **Thus, the correct chronological sequence is: Asteroid Belt → Heliopause → Oort Cloud (I - III - II).**



Source:) https://egyankosh.ac.in/bitstream/123456789/19465/1/Unit-6.pdf Pg: 28
https://egyankosh.ac.in/bitstream/123456789/59073/1/Block-1.pdf Pg: 15
https://www.bbc.com/future/article/20200908-the-weird-space-that-lies-outside-our-solar-system
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.16) Consider the following statements:
I. The Earth's orbital speed is faster at perihelion than at aphelion.
II. The region between the Tropic of Cancer and the Tropic of Capricorn is the only zone on Earth where the Sun will ever be directly overhead.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: According to **Kepler's Second Law of Planetary Motion** (the Law of Equal Areas), a planet moves faster when it is closer to the Sun and slower when it is further away. **Perihelion** is the point in Earth's orbit where it is closest to the Sun (approx. 147 million km), occurring around January 3rd. Consequently, the **Earth's orbital speed is at its maximum at perihelion and minimum at Aphelion** (approx. 152 million km), which occurs around July 4th.

As the magnitude of the angular momentum (Mass x radius x velocity) at both positions will be the same and the mass is the same. So velocity will be inversely proportional to radius. Now, as Aphelion is farther than Perihelion, **the velocity at Perihelion will be more than at Aphelion**.

Statement II is correct: The Torrid (Tropical) Zone between the Tropic of Cancer and Tropic of Capricorn is the only region where the Sun can be directly overhead. This happens due to Earth's 23.5° axial tilt, which shifts the Sun's vertical rays between 23.5°N and 23.5°S during its revolution. Hence, only this zone experiences overhead Sun, while others do not.

Source:) https://www.ncert.nic.in/textbook/pdf/keph107.pdf (Page 129)
https://egyankosh.ac.in/bitstream/123456789/32496/1/Unit-1.pdf (page 7-8)
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.17) With reference to the Convection Current Theory proposed by Arthur Holmes, which of the following is the primary source of heat that drives convection cells within the Earth's mantle?
a) Residual primordial heat retained since the Earth's formation.
b) Intense pressure generated by the weight of overlying crustal rocks.
c) Radioactive decay of elements such as uranium and thorium within the mantle.
d) Frictional heat produced by tectonic plate movement along fault zones.

Ans) c
Exp) Option c is the correct answer.
The **Convection Current Theory**, proposed by Arthur Holmes, explains that **slow-moving convection currents within the Earth's mantle drive the movement of tectonic plates**. Hot mantle material rises, spreads laterally, cools, and then sinks, creating a continuous circulation system that moves the crust.
Option c is correct: Holmes identified **radioactive decay of elements** (uranium, thorium, potassium) within the mantle as the **primary source of heat** driving these convection currents. The continuous release of heat from radioactive decay creates temperature gradients necessary for mantle circulation.
Source:) Fundamentals of Physical Geography(Class 11th-Old NCERT) Chapter-4: Distribution of Ocean and Continents Page: 31
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.18) Consider the following information:

Volcano	Feature	Country
I. Mount Etna	Active Volcano	Italy
II. Mount Taftan	Extinct Volcano	Afghanistan
III. Mount Fuji	Dormant Volcano	South Korea

In how many of the above rows is the information correctly matched?
a) Only one
b) Only two
c) All the three
d) None

Ans) a
Exp) Option a is the correct answer.
Row I is correctly matched: Located on the east coast of Sicily (Italy), **Mount Etna is one of the world's most active stratovolcanoes** and is in a near-constant state of activity.

Row II is incorrectly matched: Situated in the **Sistan and Baluchestan province of Iran, Mount Taftan** is a highly eroded polygenetic stratovolcano. While it is currently **classified as dormant** because it has not had a major eruption in recent recorded history, it still exhibits heavy fumarolic activity, emitting volcanic gases and steam from its peaks.
Row III is incorrectly matched: Mount Fuji is a stratovolcano situated on Honshu Island, Japan. **Mount Fuji is in no danger of erupting soon, but it is an 'active' volcano, so it is capable of erupting sometime in the future.** The last eruption happened in 1707 but has the potential to erupt in the future.
Source:) Certificate Physical and Human Geography by G. C. Leong, chapter 3,page 33, 30,
https://indianexpress.com/article/explained/everyday-explainers/japan-ai-video-mount-fuji-volcano-eruption-10216904/
https://egyankosh.ac.in/bitstream/123456789/98364/3/Unit%206.pdf p-115
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.19) Consider the following statements with reference to the Earth's crust:
I. Continental crust is primarily composed of granite while oceanic crust is primarily composed of basalt rocks.
II. Continental crust and Oceanic crust are separated by Mohorovicic Discontinuity.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) a
Exp) Option a is the correct answer.
The topmost layer of the Earth is the crust. The crust is formed by a combination of several rocks and minerals, and there are several layers in it. All of these layers are together known as the 'lithosphere'. The thickness of the crust may vary from a few kilometres under the oceans to 75 kilometres under the mountainous regions such as the Himalayas.
Statement I is correct: Continental crust is primarily composed of light-colored, less dense rocks like **granite,** while **oceanic crust is primarily composed of** dark, dense rocks like **basalt.**

- The **continental crust is less dense than all the layers of the Earth's interior.** Its density is 2.7g/cm^3. The main mineral composition of continental crust is **Silica and Aluminium**. This layer is also commonly known as Sial layer.
- **Oceanic crusts are heavier** and its density is 3.0 g/cm^3. It is a continuous zone of denser basaltic rocks forming the ocean floors, comprising mainly silica, and Magnesium.

Statement II is incorrect: Mohorovicic (Moho) Discontinuity is the line that separates crust from the mantle. It was discovered in 1909 by A. Mohorovicic which marks the boundary between the lower crust and the upper mantle.
The **boundary between the upper crust and the lower crust is known as the Conrad discontinuity.**
Source:) Certificate Physical and Human Geography, G.C. Leong, Chapter - 02, The Earth's Crust, Pg. 12
https://egyankosh.ac.in/bitstream/123456789/57480/1/Unit5.pdf (Pg. 9 - 10)
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust
Subtopic:)

Q.20) With reference to the Doctrine of Clean Slate under the Insolvency and Bankruptcy framework, which of the following best describes the doctrine?
a) It allows liquidation of all assets of an insolvent company without any restructuring attempt.
b) It permits a resolved company to restart operations free from past liabilities, debts, and penalties after an approved resolution plan.
c) It grants absolute immunity to former promoters and directors from all civil and criminal proceedings.
d) It enables government agencies to reopen settled claims after resolution approval.

Ans) b

Exp) Option b is the correct answer.

The **Doctrine of Clean Slate** under the Insolvency and Bankruptcy framework ensures that once a company is successfully resolved, it can function afresh without past debts, liabilities, and penalties burden, helping attract investors and enable revival.

Source: SFG Essential, Current Affairs, Quarterly Compilation (October–December 2025), Page: 18

Subject:) Current Affairs

Topic:) Polity

Subtopic:)

Q.21) Consider the following statements:

1. The Earth's magnetic field has reversed every few hundred thousand years.
2. When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.
3. When living organisms originated, they modified the early atmosphere of the Earth.

Which of the statements given above is/are correct?

a) 1 only

b) 2 and 3 only

c) 1 and 3 only

d) 1, 2 and 3

Ans) c

Exp) Option c is the correct answer.

Statement 1 is correct. The study of paleomagnetism (the magnetism residual in rocks) has provided evidence that **the earth's magnetic field has reversed direction every few hundred thousand years**– i.e., the north and south poles have switched polarity. Some scientists predict this phenomenon may occur in the near geologic future.

Statement 2 is incorrect. **When Earth was first created, there was no oxygen whatsoever**. Our Earth's atmosphere at that time consisted mostly of methane and carbon dioxide. The early atmosphere largely contained water vapour, nitrogen, carbon dioxide, methane, and ammonia. These gases were released from the interior solid earth during its cooling off. The process through which the gases were outpoured from the interior is called degassing.

Statement 3 is correct. The process of photosynthesis modified the early atmosphere of the Earth. About 2.4 billion years ago, a type of organism called cyanobacteria evolved on the early Earth and began carrying out photosynthesis. Photosynthesis uses carbon dioxide and energy from the Sun to produce sugar and oxygen. The cyanobacteria were very simple organisms but performed an important role in changing Earth's early atmosphere. They carried out photosynthesis to produce the materials they needed to grow. They gave off oxygen to the atmosphere as they did this.

Source: UPSC CSE Pre 2018

Subject:) Geography

Topic:) The Earth and the Universe

Subtopic:)

Q.22) Which of the following best describes the geological feature known as a 'Lacolith'?

a) A vertical or near-vertical wall-like intrusive body that cuts across the layers of the surrounding country rock.

b) A large, dome-shaped intrusive body connected by a pipe-like conduit from below.

c) A saucer-shaped intrusive body that develops a concave-upward shape due to the weight of the magma.

d) A lens-shaped mass of igneous rock that occupies the crest of an anticline or the trough of a syncline in folded strata.

Ans) b

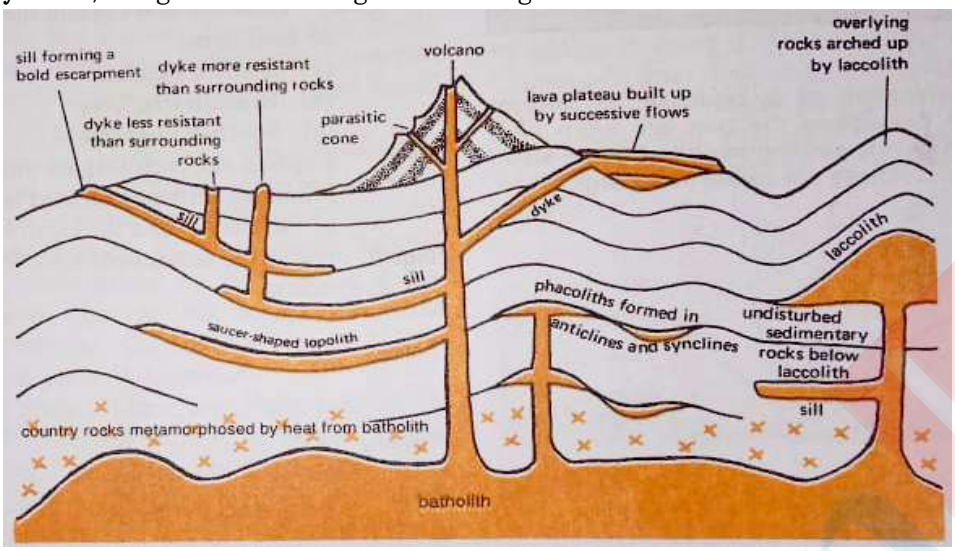
Exp) Option b is the correct answer.

Option a is incorrect. A **dyke (or dike)** is a **vertical or steeply inclined** sheet of igneous rock that cuts **across existing rock layers.**

Option b is correct. A **laccolith** forms when magma intrudes between rock layers and **pushes the overlying strata upward into a dome shape**. It is often fed by a **pipe-like conduit** through which magma rises from deeper layers. This gives it a characteristic **mushroom or dome-like appearance.**

Option c is incorrect. A **lopolith** is a large, **basin-shaped** intrusion where the central part sinks downward due to the weight of accumulated magma, forming a saucer-like structure.

Option d is incorrect. A **phacolith** forms along the axes of folds either at the **crest of an anticline or the trough of a syncline,** filling these folded regions with magma.



Source:) GC Leong- Chapter 3-Vulcanism and Earthquake - Page-20
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

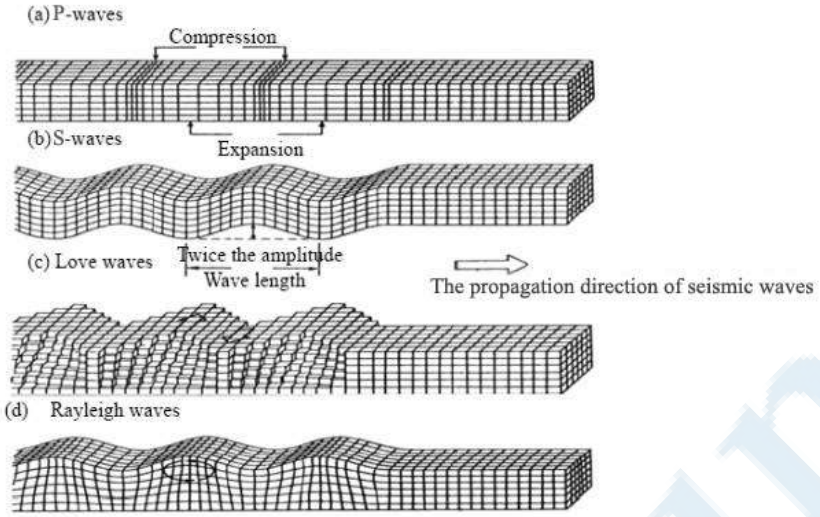
Q.23) Although the Sun is a massive celestial body, it is generally believed that it will never evolve into a Black Hole. Which of the following is the primary reason for this?
a) Sun has a high concentration of heavy metals at the core.
b) Sun lacks sufficient mass to undergo a supernova explosion.
c) The core temperature of the Sun is too high.
d) Sun has weak magnetic fields.

Ans) b
Exp) Option b is the correct answer.
A star must be very massive to form a black hole after a core-collapse supernova. **The Sun is a low-mass star and does not have enough mass to explode as a supernova or become a black hole.** It will instead become a red giant and finally a white dwarf.
"Heavy metals" affect how a star burns and its lifespan, but they do not prevent a gravitational collapse if the mass is sufficient.
A high core temperature prevents collapse by providing outward pressure. However, a black hole forms specifically when the temperature drops.
Source:) https://www.nasa.gov/image-article/why-sun-wont-become-black-hole/
https://egyankosh.ac.in/bitstream/123456789/12210/1/Unit-10.pdf Pg: 30, 31
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.24) With reference to Earthquakes, consider the following statements about P-waves or the Primary Waves:
I. These are the fastest seismic waves and are the first to be recorded on a seismograph.
II. In P-waves, individual particles vibrate parallel to the direction of wave propagation.

III. They can travel only through solid material and cannot pass through liquid or gaseous material.
IV. They are the most destructive type of seismic waves and cause maximum damage on the Earth's surface.
Which of the statements given above is/are correct?
a) I, II, and IV only
b) I and II only
c) I only
d) II, III, and IV only

Ans) b
Exp) Option b is the correct answer.
Earthquakes generate different types of seismic waves that travel through the Earth. Earthquake waves are basically of two types- body waves and surface waves. The body waves are further divided into P and S-waves.



Option I is correct. P-wave**s travel at the highest speed among all seismic wave**s, so they reach seismograph stations before other waves like S-waves and surface waves.
Option II is correct. In P-waves, particles move back and forth in the same direction as the wave travels i.e. the particle **vibrates parallel to the direction of wave propagation**. This type of motion is called longitudinal motion and is similar to how sound waves move through air.
Option III is incorrect. P-waves can travel through **solid, liquid, and gaseous material.** This property distinguishes them from S-waves, which can travel only through solids.
Option IV is incorrect. P-waves are generally th**e least destructive**. The most destructive waves are surface waves, which cause strong ground shaking and damage structures.
Source:) NCERT-Class 11 (World Physical geography)- chapter-3-INTERIOR OF THE EARTH, page 23
https://egyankosh.ac.in/bitstream/123456789/70609/1/Unit-2.pdf
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.25) Consider the following:
1. Pyroclastic debris
2. Ash and dust
3. Nitrogen compounds
4. Sulphur compounds
How many of the above are products of volcanic eruptions?
a) Only one
b) Only two
c) Only three
d) All four

Ans) d

Exp) option d is the correct answer.

A volcano is a place where gases, ashes and/or molten rock material – lava – escape to the ground. The material that reaches the ground includes lava flows, **pyroclastic debris,** volcanic bombs, **ash and dust** and gases such as **nitrogen compounds**, **Sulphur compounds** and minor amounts of chlorene, hydrogen and argon.

Source: UPSC CSE Pre 2024

Subject:) Geography

Topic:) Vulcanism and Earthquakes

Subtopic:)

Q.26) Consider the following statements:

Statement I: The shallow focus earthquake generally causes more damage than a deep focus earthquake.

Statement II: Compared to a deep focus earthquake, seismic waves from a shallow focus earthquake reach the Earth's surface with greater intensity due to lesser loss of energy.

Which of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I.

b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.

c) Statement I is correct but Statement II is not correct.

d) Statement I is not correct but Statement II is correct.

Ans) a

Exp) Option a is the correct answer.

Earthquake impact depends on both magnitude and depth, as depth affects how much energy reaches the surface and influences the level of damage.

Statement I is correct. Shallow-focus earthquakes are actually **more destructive**. They originate close to the Earth's surface, usually within 0–70 km depth. Because of this proximity, the seismic waves hit the surface with strong force. This leads to intense ground shaking. Such earthquakes often result in **severe damage to buildings, infrastructure, and higher casualties**. The recent Kamchatka earthquake is a good example of destruction caused by shallow earthquakes.

Statement II is correct and explains Statement I. When an earthquake occurs at shallow depth, the seismic waves travel a shorter distance. Due to this, **energy loss during propagation is minimal**. As a result, the waves retain high intensity when they reach the surface.

In contrast, deep earthquakes lose significant energy while travelling long distances. Hence, their surface impact is weaker.

Source: https://www.livemint.com/news/world/why-was-russias-8-8-magnitude-earthquake-destructive-and-deadly-all-to-know-about-shallow-and-deep-intensity-tremors-11753846780030.html

https://www.thehindu.com/news/international/Difference-between-shallow-and-deep-earthquakes/article60494113.ece

https://egyankosh.ac.in/bitstream/123456789/70609/1/Unit-2.pdf

Subject:) Geography

Topic:) Vulcanism and Earthquakes

Subtopic:)

Q.27) Consider the following statements with respect to the process of Solifluction:

I. It is a type of mass movement that involves the gradual downslope movement of water-saturated fine grained rock debris.

II. It does not occur in permafrost regions.

Which of the statements given above is/are correct?

a) I only

b) II only

c) Both I and II

d) Neither I nor II

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: Solifluction is a type of mass movement characterized by the slow downslope flow of water-saturated or lubricated soil mass and fine-grained rock debris. This phenomenon is essentially a form of creep, where the movement is facilitated by the presence of lubricating water. It is prominently observed in permafrost zones, where groundwater strata are situated between permanently frozen soil and rocks, contributing to the gradual downslope motion typical of solifluction.

Statement II is incorrect: Solifluction is most commonly associated with permafrost regions, where the active layer thaws in summer and becomes waterlogged above frozen ground, facilitating movement of soil.

Source:) https://egyankosh.ac.in/bitstream/123456789/98372/3/Unit%208.pdf Pg: 174

Subject:) Geography

Topic:) Weathering, Mass Movements, and Groundwater

Subtopic:)

Q.28) Large mineral grains formed in igneous rocks indicate which one of the following?

a) Rapid cooling of magma at the Earth's surface.

b) High silica content in the magma during crystallization.

c) Slow cooling of magma beneath the Earth's surface.

d) Release of volcanic gases through fissures and vents.

Ans) c

Exp) Option c is the correct answer.

Igneous rocks are formed by the cooling and solidification of molten rock (magma) from beneath the Earth's crust. They are normally crystalline in structure. They do not occur in strata (layers) nor do they contain fossils.

Option c is correct: Large mineral grains in an igneous rock indicate slow cooling of magma, typically occurring deep within the Earth's crust. This slow cooling process gives crystals ample time to grow, often resulting in an intrusive or plutonic, coarse-grained texture where crystals are visible to the naked eye. **Plutonic rocks** are igneous rocks, formed at some depth in the earth's crust. They **have cooled and solidified slowly so that large, easily-recognized crystals have been able to form.**

Source:) Certificate Physical and Human Geography, G.C. Leong, Chapter - 02, The Earth's Crust, Pg. 13

Subject:) Geography

Topic:) Geomorphology: The Earth's Crust

Subtopic:)

Q.29) With reference to the International Date Line (IDL), consider the following statements:

I. When one crosses the International Date Line from west to east, he/she loses a day.

II. The International Date Line has no formal treaty-based legal status.

Which of the statements given above is/are correct?

a) I only

b) II only

c) Both I and II

d) Neither I nor II

Ans) b

Exp) Option b is the correct answer.

The **International Date Line (IDL)** is an imaginary line located roughly along 180° longitude, exactly opposite to the **Prime Meridian (0° longitude).** It lies halfway around the Earth from Greenwich. The time at this line differs by 12 hours from the Prime Meridian. Although it broadly follows 180°, it is not a straight line and is often referred to as the **'Line of Demarcation'** because it separates one calendar day from the next.

Statement I is incorrect. **When one crosses the International Date Line from west to east (or moving eastward or moving eastbound), he/she gains a day (because of the gain in time he encountered).** Conversely, when crossing

from east to west (or moving westward or moving westbound), he/she loses a day (because of the loss in time he has made). This is done to adjust the date because the Earth is divided into 24 time zones and rotates once every 24 hours. The IDL serves as the point where the date changes by exactly one day.
When a person crosses the IDL while travelling eastward, the calendar moves backward. This means the date is subtracted by one day. In practical terms, the traveller **gains a day**.
Statement II is correct. The **International Date Line has no formal treaty-based legal status**, so countries can decide their own time zones and calendar dates.
Source: https://egyankosh.ac.in/bitstream/123456789/80717/3/Unit-3.pdf - Time Calculation - International Date Line (IDL) – Page 46,47
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust
Subtopic:)

Q.30) The substances Mephedrone, Amphetamines, and MDMA, often seen in the news, are classified as which of the following?
a) Naturally occurring opioid substances derived from poppy latex.
b) Pesticides banned under the Rotterdam Convention.
c) Persistent Organic Pollutants (POPs) listed under the Stockholm Convention.
d) Synthetic psychotropic stimulant drugs.

Ans) d
Exp) Option d is the correct answer.
Mephedrone, amphetamines, and MDMA are **synthetic psychotropic stimulants** that alter mood, cognition, and behaviour. It is classified as New Psychoactive Substance (NPS).
Source: SFG Essential, Current Affairs, Quarterly Compilation (October–December 2025), Page: 48
Subject:) Current Affairs
Topic:) Security
Subtopic:)

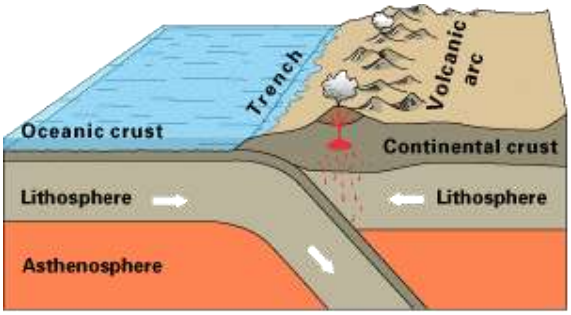
Q.31) The tail of comet is directed away from the Sun, because–
a) As the Comet rotates around the Sun, the lighter mass of the Comet is pushed away due to centrifugal force alone.
b) As the Comet rotates, the lighter mass of the Comet is attracted by some stars situated in the direction of its tail.
c) The radiation emitted by the Sun exerts a radial pressure on the comet throwing its tail away from the Sun.
d) The Tail of the Comet always exists in the same orientation.

Ans) c
Exp) Option c is the correct answer.
The radiation emitted by the Sun exerts a radial pressure on the comet throwing its tail away from the Sun. This is because the **comet's tail is composed of dust and gas that are released from the comet's nucleus by the Sun's heat and driven away by sunlight** (solar radiation pressure) and the solar wind. The solar wind is a stream of charged particles that flows outward from the Sun at high speeds. The solar wind interacts with the gas in the comet's tail, causing it to glow and point along the streamlines of the solar wind. The tail always points away from the Sun, whether the comet is approaching or receding from it.
Source: UPSC CSE Pre 1997
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.32) Consider the following landforms/features:
I. Oceanic trenches
II. Volcanic Arc
III. Rift valleys

IV. Mid-oceanic ridges
V. Island Arc
How many of the above are primarily formed at divergent plate boundaries?
a) Only two
b) Only three
c) Only four
d) All the five

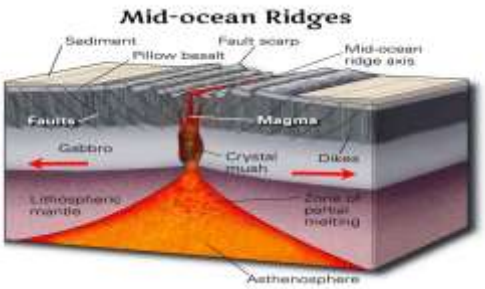
Ans) a
Exp) Option a is the correct answer.
Plate tectonics is a scientific theory that explains how major landforms are created as a result of Earth's subterranean movements.
Option I is incorrect. Oceanic trenches are deep underwater depressions **that form at convergent plate boundaries**, where two tectonic plates collide. In this collision, one of the plates, typically a denser oceanic plate, is forced beneath the other plate, leading to the formation of a trench. These trenches are the deepest parts of the ocean and often associated with intense seismic activity and volcanic eruptions.
Option II is incorrect. A **volcanic arc forms at convergent boundaries,** particularly where an oceanic plate subducts beneath a continental plate. The melting of the subducted plate generates magma, which rises to form **volcanoes in an arc shape**.



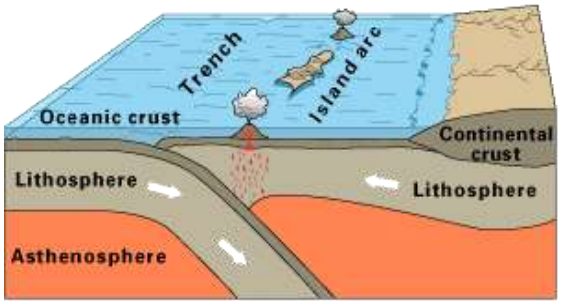
Oceanic-continental convergence

Option III is correct. Rift valleys are a feature of **divergent plate boundaries**. They form when tectonic plates move apart, causing the crust to stretch, thin, and eventually crack. The central block subsides, forming a valley. The **East African Rift Valle**y is a well-known example.

Option IV is correct. Mid-ocean ridges are also formed at **divergent boundaries,** where oceanic plates move apart and magma rises from below to create new crust. These ridges are underwater mountain chains and represent active zones of seafloor spreading.



Option V is incorrect. Island arcs are formed at **convergent boundaries** involving **oceanic-oceanic plate subduction.** The subducting plate melts and produces magma, leading to volcanic islands arranged in an arc shape, such as Japan.



Oceanic-oceanic convergence

Source:) https://egyankosh.ac.in/bitstream/123456789/83716/1/Unit-15.pdf
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.33) Consider the following discontinuities found within the Earth's interior:
I. Gutenberg Discontinuity
II. Lehmann Discontinuity
III. Conrad Discontinuity
IV. Repetti Discontinuity
Which one of the following is the correct sequence of the above discontinuities as they occur from the Earth's Crust to the Core?
a) III – IV – I – II
b) III – IV – II – I
c) IV – III – I – II
d) IV – III – II – I

Ans) a
Exp) Option a is the correct answer.
III. Conrad Discontinuity: It is the uppermost boundary located within the Earth's crust, **separating the outer granitic (Sial) layer from the inner basaltic (Sima) layer.**
IV. Repetti Discontinuity: Located at a **depth of about 660–700 km**, this zone marks the transition between the **Upper Mantle (Asthenosphere) and the Lower Mantle (Mesosphere).**
I. Gutenberg Discontinuity: Also known as the Core-Mantle Boundary (CMB), it **separates the solid lower mantle from the liquid outer core.** (At depth of 2,900 km)
II. Lehmann Discontinuity: This is the deepest boundary, **separating the liquid iron-nickel outer core from the solid iron-nickel inner core.**
Source:)
Fundamentals of Physical Geography, class 11th NCERT, Chapter 3rd, page22, 23
https://egyankosh.ac.in/bitstream/123456789/83703/1/Unit-10.pdf page 14,15,16
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust

Subtopic:)

Q.34) The maximum mass that a stable white dwarf can have is known as:
a) The Roche Limit
b) The Chandrasekhar Limit
c) The Armstrong Limit
d) The Eddington Limit

Ans) b
Exp) Option b is the correct answer.
'The Chandrasekhar Limit' – explained the maximum mass of a stable white dwarf star.
Stars can end their lives as slowly cooling stellar objects known as white dwarfs or they can go on for supernova explosions.
The Chandrasekhar limit determines whether a star ends its life as a smoldering white dwarf, or explodes in a supernova to become a neutron star or black hole. The Chandrasekhar value for a white dwarf star is generally considered to be 1.4 times the mass of the Sun.
White dwarfs are stars that have burned up all of the hydrogen they once used as nuclear fuel.
Option a is incorrect: Roche Limit refers to the distance within which a celestial body, held together only by its own gravity, will disintegrate due to a second celestial body's tidal forces.
Option c is incorrect: The Armstrong Limit refers to the altitude (approx. 19 km) where atmospheric pressure is so low that water (and blood) boils at human body temperature.
Option d is incorrect: Eddington Luminosity/Limit is the maximum luminosity a stellar body can achieve when there is a balance between the force of radiation acting outward and the gravitational force acting inward.
Source:) https://indianexpress.com/article/technology/science/will-our-universe-rip-apart-or-expand-forever-dark-energy-holds-the-answer-10198693/
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.35) With reference to the writ of Habeas Corpus in the Indian judiciary, consider the following statements:
I. It cannot be issued against a private individual.
II. If detention is found illegal, the court can order the immediate release of the detained person.
III. A Habeas Corpus petition cannot be entertained once a judicial remand order has been passed.
IV. It can be issued by both the Supreme Court and High Courts.
Which of the above statements given above are correct?
a) I and III only
b) II and IV only
c) II, III and IV
d) I, II and IV

Ans) b
Exp) Option b is the correct answer.
Habeas Corpus is one of the most important constitutional writs for protecting personal liberty. The Latin term means "to have the body." It is a judicial order directing the authority or person detaining someone to produce the detained individual before the court.
Statement I is incorrect. Habeas Corpus is not restricted to government or public authorities. It **can also be issued against private individuals** if they have unlawfully confined or detained a person.
Statement II is correct. The central purpose of Habeas Corpus is to provide quick judicial review of detention. **Once the court determines that the detention has no legal basis, it can order the immediate release of the person to restore their liberty.**

Statement III is incorrect. Even after a judicial remand order, a Habeas Corpus petition may still be entertained in specific circumstances for example, if the remand itself is without jurisdiction or legally invalid or wholesale violation of fundamental rights.

Statement IV is correct. The Supreme Court can issue Habeas Corpus under Article 32 for enforcement of Fundamental Rights, and High Courts can issue it under Article 226, which has even wider jurisdiction.

Source: SFG Essential, Current Affairs, Quarterly Compilation (October–December 2025), Page: 3
Subject:) Current Affairs
Topic:) Polity
Subtopic:)

Q.36) Consider the following statements:

Statement I: In a hemisphere, the time gap between the summer solstice and the winter solstice is approximately six months.

Statement II: The Earth revolves around the Sun in an elliptical orbit, causing the distance between the Earth and the Sun to vary throughout the year.

Which one of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I
b) Both Statement I and Statement II are correct, but Statement II does not explain Statement I
c) Statement I is correct but Statement II is not correct
d) Statement I is not correct but Statement II is correct

Ans) b

Exp) Option b is the correct answer.

Statement I is correct: In any given hemisphere, the **longest day** occurs during the **Summer Solstice** (around June 21 in the Northern Hemisphere) and the **shortest day** occurs during the **Winter Solstice** (around December 22 in the Northern Hemisphere). The time elapsed between these two points is approximately **six months**. This cycle is a fundamental characteristic of the Earth's annual journey around the Sun.

Statement II is correct: The Earth does revolve in an **elliptical orbit**, leading to a varying distance from the Sun (~147 million km at **Perihelion** to ~152 million km at **Aphelion**).

Statement II is not the correct explanation for Statement I. The **variation in day length and the six-month gap between the longest and shortest days are caused by the 23.5-degree axial tilt** of the Earth being maintained in a constant direction as it revolves. If the Earth's orbit were a perfect circle (constant distance), we would still have a six-month gap between the longest and shortest days because of the tilt. The elliptical orbit affects the intensity of solar radiation and the speed of the Earth in its orbit, but not the fundamental occurrence of the solstices.

Source:) https://ncert.nic.in/textbook/pdf/gecu112.pdf (page 177-179)
https://egyankosh.ac.in/bitstream/123456789/83534/3/Unit-3.pdf (page 55-56)
https://egyankosh.ac.in/bitstream/123456789/59073/1/Block-1.pdf (page 12)
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.37) Which of the following mountain ranges are formed due to oceanic-continental plate convergence?

I. Himalayas
II. Andes
III. Alps
IV. Atlas

Select the correct answer using the code given below:

a) II only
b) I and III only
c) II and IV only
d) I, II and IV

Ans) a

Exp) Option a is the correct answer.

Fold mountains are formed as a result of **compressional forces generated at convergent plate boundaries**, where two tectonic plates move towards each other, leading to the **folding, crumpling, and upliftment of sedimentary rock strata**. There are three types of fold mountains based on type of plates involved in collision:

- **Continental–Continental Collision:** Fold mountains formed due to the collision of two continental plates are characterized by intense folding and crustal thickening eg. HImalayas Alps and Atlas mountains
- **Continental–Oceanic Collision:** Fold mountains formed due to the convergence of a continental plate and an oceanic plate involve the subduction of the denser oceanic plate beneath the continental plate, resulting in folding and volcanic activity along the continental margin; for example, the Andes Mountains
- **Oceanic–Oceanic Collision:** Fold mountains formed due to the convergence of two oceanic plates involve the subduction of one oceanic plate beneath the other, leading to the formation of folded structures and volcanic island arcs; for example, the Japanese Archipelago has been formed due to such oceanic–oceanic convergence.

Option I is incorrect: The Himalayas are the product of **continental-continental plate convergence** – specifically the collision of the Indian Plate with the Eurasian Plate. Both are continental plates; neither is oceanic.

Option II is correct: The Andes are an example of **oceanic-continental convergence. The Nazca Plate (oceanic) subducts beneath the South American Plate (continental)** along the Peru-Chile Trench.

Option III is incorrect: The Alps are formed by the convergence of the African Plate and the Eurasian Plate – **both continental plates**.

Option IV is incorrect: The Atlas Mountains of northwestern Africa are primarily formed through the convergence of the African and Eurasian plates. This is a **continental-continental interaction.**

Source:) https://egyankosh.ac.in/bitstream/123456789/53275/3/Block-4.pdf P-54

Subject:) Geography

Topic:) Vulcanism and Earthquakes

Subtopic:)

Q.38) With reference to the Earth's Inner Core, consider the following statements:

I. It is primarily composed of iron and nickel.

II. It remains solid despite extremely high temperatures due to immense pressure at the Earth's center.

III. It rotates at a faster rate and in the opposite direction to the rest of the Earth.

Which of the statements given above are correct?

a) I and III only

b) I and II only

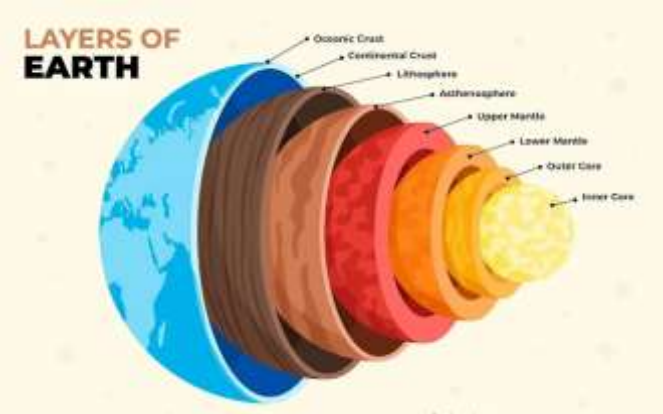
c) II and III only

d) I, II and III

Ans) b

Exp) Option b is the correct answer.

The Earth is made up of different layers namely, the crust, mantle, outer core, and inner core. The inner core is the deepest part of the Earth and lies at its center.



Statement I is correct. The **inner core is mainly made up of heavy metals, especially iron and nickel.** These materials are similar to those found in meteorites, which suggests that the Earth's core has a metallic composition.

Statement II is correct. The temperature in the inner core is extremely high but the pressure at the Earth's center is so intense that it prevents the material from melting. Because of this **immense pressure,** the **inner core remains solid even under such extreme heat.**

Statement III is incorrect. Recent studies confirm Earth's inner core operates on a 60–70 year cycle, **alternating between rotating faster and slower than the surface**. While often experiencing "super-rotation" (spinning faster), data from the recent research indicates the inner core's rotation began a significant deceleration around 2010, making it temporarily slower than the surface.

Also, **it rotates in the same direction to the rest of the Earth.**

Source:) Savinder Singh-chapter-Constitution of Earth's Interior- page-109-113

https://www.thehindu.com/sci-tech/science/earths-inner-core-rotating-slower-than-surface-study-suggests/article66432473.ece

Subject:) Geography

Topic:) Geomorphology: The Earth's Crust

Subtopic:)

Q.39) Consider the following:

I. Different places on Earth experience sunrise and sunset at different times.

II. The shape of the Moon appears to be changing each night.

III. Ships disappear gradually over the horizon, with the lower part vanishing first.

IV. If poles of equal height are placed at regular intervals on level ground, their tops lie along a straight horizontal line.

How many of the above indicate that the Earth is spherical in shape?

a) Only one

b) Only two

c) Only three

d) All the four

Ans) b

Exp) Option b is the correct answer.

The Earth is not perfectly round; it is slightly flat at the poles and bulged at the equator, the shape called a geoid.

Statement I is correct. The Earth rotates from **west to east**. Because of the **curvature of the Earth**, sunlight reaches different places at different times. As a result, places located in the **eastern direction experience sunrise earlier**, while places in the **western direction experience it later**. If the Earth were flat, **all places would experience sunrise and sunset simultaneously.**

Statement II is incorrect. The **moon appears to change shape each night. The different shapes of the Moon that we see at different times of the month are called the Moon's phases.**

The Moon does not produce its own light. We are able to see the Moon due to the Sun's light illuminating the Moon. On Earth, our view of the illuminated part of the Moon changes each night, depending on where the Moon is in its orbit, or path, around Earth. Thus, the phases of the Moon are caused by the changing positions of the Moon around

the Earth. As the Moon goes around the Earth, different parts of it are illuminated by the Sun. But because the same side of the Moon always points toward Earth, most of the time some of the illuminated part (daytime on the Moon) faces away from us. And, some of the non-illuminated part (nighttime on the Moon) faces toward us. This creates the different phases of the Moon.

Statement III is correct. When a ship moves away from the observer, it does not disappear all at once. The **lower part of the ship disappears first**, followed by the upper part. This happens due to the **curved surface of the Earth**, which gradually hides the object from view. If the Earth were flat, the ship would simply appear **smaller and smaller**, but it would remain fully visible.

Statement IV is incorrect. When poles of equal height are placed at regular intervals, they do not form a perfectly straight line. The **central pole appears slightly higher** than the others. This happens because the Earth's surface is **curved and not flat**.

Source: Geography Redbook by ForumIAS 3rd Edition - Chapter 1 - The Earth - Evidence of Earth's Sphericity - Page 4
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust
Subtopic:)

Q.40) Which of the following is correct about the Central Pay Commission of India?
a) It is a statutory body.
b) It operates under the Department of Expenditure, Ministry of Finance.
c) The first Pay Commission was constituted in 1950.
d) The recommendations made by the Commission are binding on the Union Government.

Ans) b
Exp) Option b is the correct answer.
Option a is incorrect. The Central Pay Commission is **not a statutory body**. It is constituted every ten years by the **executive decision of the Union Government**, not through a law passed by Parliament.
Option b is correct. The Pay Commission functions under the **Department of Expenditure** in the Ministry of Finance.
Option c is incorrect. The **First Central Pay Commission was constituted in 1946**, even before Independence, and it submitted its report in 1947.
Option d is incorrect. The Pay Commission is **only an advisory body**. Its recommendations are not binding. The Union Government has full discretion to accept, modify, or reject any recommendation.
Source: SFG Essential, Current Affairs, Quarterly Compilation (October–December 2025), Page: 9
Subject:) Current Affairs
Topic:) Polity
Subtopic:)

Q.41) Which one of the following is a permeable rock that allows water to pass through it?
a) Granite
b) Limestone
c) Quartzite
d) Sill

Ans) b
Exp) Option b is the correct answer
The flow of water through the rocks depends on the permeability of the rock layers. In rocks, permeability depends on the size of the pores and on how connected they are to one another. **Rocks, such as limestone are porous,** and **they also crack very easily allowing the water to travel through.** Limestone rocks are full of tiny pores. These spaces have an enormous ability to carry and hold water.
Source: UPSC CAPF 2023
Subject:) Geography
Topic:) Geomorphology: The Earth's Crust

Subtopic:)

Q.42) Consider the following statements:
Statement I: The duration of dawn and twilight is longer at the poles compared to the equatorial regions.
Statement II: The linear velocity of Earth's rotation is highest at the equator and decreases towards the poles.
Which one of the following is correct in respect of the above statements?
a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is not correct.
d) Statement I is not correct but Statement II is correct.

Ans) b
Exp) Option b is the correct answer.
Statement I is correct: At the poles, the sun's path is highly oblique (tilted) to the horizon. Therefore, **the sun takes a much longer time to pass through the 18-degree zone below the horizon that defines twilight.** At the equator, the sun rises and sets almost vertically, making the transition very short.
Statement II is correct: The linear velocity of Earth's rotation is actually **highest at the equator (approx. 1670 km/hr) and decreases to zero at the poles.**
Statement II does not explain Statement I: The duration of dawn and twilight is primarily because of the oblique (slanted) angle at which the sun crosses the horizon at high latitudes, compared to the almost vertical path at the equator.
The rotational speed of Earth over the Equator and Poles does not influence the duration of dawn and twilight.
Source:) Certificate Physical and Human Geography by G. C. Leong, Chapter 1, page 9
Fundamentals of Physical Geography (NCERT), chapter 2
Earth rotates at 1,670 km per hour, so why don't we feel it? - India Today
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.43) With reference to 'Mud volcanoes', consider the following statements:
I. These are geological formations that release mud, water, and gases instead of molten lava.
II. India's only active mud volcano is located at Baratang Island in the Andaman and Nicobar Islands.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer.
Not all volcanic landforms are formed by molten lava. Some are created by the release of gases, mud, and other semi-liquid materials. Mud volcanoes are one such unique geomorphic feature.
Statement I is correct. Mud volcanoes are different from conventional volcanoes. They **do not erupt molten lav**a. Instead, they release **mud, water, gases, and fine sediments**. These materials are pushed upward due to pressure from gases trapped below the Earth's surface.
Statement II is correct. **India's only known active mud volcano is situated at Baratang Island** in the Andaman and Nicobar Islands. This region lies in a tectonically active zone. The mud volcano here has shown periodic activity, with eruptions of mud and gases. Recent Geological studies, including those by GSI, indicate that the material from this mud volcano is **very old (dating back to the Oligocene period)**, highlighting its geological significance.
Source: https://egyankosh.ac.in/bitstream/123456789/87672/1/Unit-5.pdf - Tectonic and Volcanic Landforms - Landforms Associated with Different Types of Materials - Mud Volcano – Page:117

https://www.thehindu.com/news/national/andaman-and-nicobar-islands/samples-of-indias-only-active-mud-volcano-in-andaman-date-back-to-23-million-years-gsi/article70182508.ece
Subject:) Geography
Topic:) Vulcanism and Earthquakes
Subtopic:)

Q.44) Due to the interaction between the Sun and an intruding star, a huge amount of matter was ejected from the Sun, and a cigar-shaped filament was created. This filament then started revolving around the Sun due to the effect of gravitation. With time, the filament broke into several pieces which eventually cooled and condensed to form the planets.
Which of the following theories/hypotheses is described above?
a) Steady State Theory
b) Tidal Hypothesis
c) Nebular Hypothesis
d) Binary Star Theory

Ans) b
Exp) Option b is the correct answer.
Option a is incorrect: Steady State Theory: A cosmological theory suggesting the universe has no beginning or end and maintains a constant density by continuous creation of matter.
Option b is correct: The description in the statement refers to the Tidal Hypothesis, which was propounded by Sir James Jeans and later modified by Harold Jeffreys in the early 20th century. According to this theory, a **massive "wandering star" approached the primitive sun**. As the star came closer, its immense gravitational pull created a "tidal surge" on the surface of the sun. This pull was so strong that it drew out a long, cigar-shaped filament of gaseous matter from the sun's outer surface. As the wandering star moved away, the filament detached from the sun. Due to the gravitational instability, this filament broke into several pieces, which eventually cooled and condensed to form the planets.
Due to the tidal effect, the filament remained thicker in the center and thinner at the ends. Hence, **the planets formed by this filament are bigger in the center and smaller at the sides.**
Option c is incorrect: Nebular Hypothesis: A rotating cloud of gas (nebula) cooled and flattened, shedding rings that formed planets.
Option d is incorrect: Binary Star Theory: The sun initially had a companion star. A third "intruding star" collided with or passed near the companion, breaking it into planetary matter.
Source:) https://egyankosh.ac.in/bitstream/123456789/57462/1/Unit1.pdf Pg: 18, 19, 20, 21, 22.
Subject:) Geography
Topic:) The Earth and the Universe
Subtopic:)

Q.45) The "Drone Wall Initiative" recently seen in the news refers to which of the following?
a) A coordinated EU–NATO defence system along Europe's Eastern flank to counter increasing drone threats, especially from Russia.
b) A maritime security initiative to deploy drones for monitoring submarine movements in the South China Sea.
c) An Indian Ministry of Defence project aimed at securing the entire length of the international border using surveillance drones.
d) An Indo-Pacific naval project deploying drone swarms for maritime surveillance.

Ans) a
Exp) Option a is the correct answer.
The Drone Wall Initiative is a security project led by the European Union in coordination with NATO partners to create an integrated counter-drone shield along its eastern flank, particularly in response to rising aerial intrusions and asymmetric drone threats from Russia. It involves AI-enabled detection systems, real-time surveillance, interception capabilities, and electronic warfare tools such as GPS jamming.

Source: SFG Essential, Current Affairs, Quarterly Compilation (October–December 2025), Page: 46
Subject:) Current Affairs
Topic:) Security
Subtopic:)

Q.46) If today is Monday, then which day of the week will it be exactly after $13^{11} + 14^{12} + 15^{13}$ days?
a) Monday
b) Wednesday
c) Thursday
d) Friday

Ans) a
Exp) Option a is the correct answer.
A week has **7 days**, so we only need the remainder on division by **7**.

- $\mathbf{14^{12}}$ is a multiple of **7**, so it contributes **0**
- $\mathbf{13^{11}}$ leaves the same remainder as $\mathbf{6^{11}}$; powers of **6** alternate **6, 1**, so odd power gives **6**
- $\mathbf{15^{13}}$ leaves the same remainder as $\mathbf{1^{13} = 1}$

So total remainder = **6 + 0 + 1 = 7**, i.e. an exact multiple of **7**.
Hence the day after $\mathbf{13^{11} + 14^{12} + 15^{13}}$ days will again be **Monday**.
Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)

Q.47) There are three traffic signals. Each signal changes colour from green to red and then from red to green. The first signal stays green for 33 seconds, the second signal stays green for 35 seconds and the third signal stays green for 65 seconds. The duration of the red colour is exactly 15 seconds for all three signals. At 5:00 p.m., they together turn green. At what exact time will they change to red next, simultaneously?
a) 5:20:00 p.m.
b) 5:19:45 p.m.
c) 5:15:00 p.m.
d) 5:25:15 p.m.

Ans) b
Exp) Option b is the correct answer.
Each signal repeats after:

- First: **33 + 15 = 48 sec**
- Second: **35 + 15 = 50 sec**
- Third: **65 + 15 = 80 sec**

They will again **turn green together** after:
LCM(48, 50, 80) = 2400 sec = 40 min
So they are together green again at:
5:40:00 p.m.
But need time when they **next turn red together**, i.e. after starting green together, each must complete its own green time.
So required time **t** must satisfy:

- **t ≡ 33 (mod 48)**
- **t ≡ 35 (mod 50)**
- **t ≡ 65 (mod 80)**

Check the options, **5:19:45 p.m. = 1185 sec after 5:00**
Now:

- **1185 − 33 = 1152**, divisible by **48**

- **1185 – 35 = 1150**, divisible by **50**
- **1185 – 65 = 1120**, divisible by **80**

Hence all three turn red together at **5:19:45 p.m.**
Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)

Q.48) What is the middle term of the sequence
A, Z, Z, B, B, B, Y, Y, Y, Y, C, C, C, C, C, . . ., N ?
a) I
b) J
c) Q
d) R

Ans) b
Exp) Option b is the correct answer.
The letters come alternately from the **beginning and end** of the alphabet:
A, Z, B, Y, C, X, D, W, ...
Also, the repetitions increase as:
A once, Z twice, B thrice, Y four times, ...
So the 26 blocks have sizes:
1, 2, 3, 4, ..., 26
Total number of terms:
1 + 2 + 3 + ... + 26 = (26 × 27)/2 = 351
Hence the middle term is the:
(351 + 1)/2 = 176th term.
Now,
1 + 2 + ... + 18 = (18 × 19)/2 = 171
1 + 2 + ... + 19 = (19 × 20)/2 = 190
So the **176th term** lies in the **19th block**.
The 19th block is: **J**
So the middle term is **J**
Subject:) CSAT
Topic:) Logical and Verbal Reasoning
Subtopic:)

Q.49) The letters of the word "SUBORDINATES" are rearranged such that the vowels are arranged alphabetically, and the consonants are arranged alphabetically, but the relative positions of the vowels and consonants remain unchanged. How many positions of the letter/letters will remain unchanged compared to the original word?
a) None
b) One
c) Two
d) Three

Ans) b
Exp) Option b is the correct answer.
Original word:
S U B O R D I N A T E S
Vowel positions: **2, 4, 7, 9, 11**
Vowels = **U, O, I, A, E** → alphabetically: **A, E, I, O, U**
Consonant positions: **1, 3, 5, 6, 8, 10, 12**

Consonants = **S, B, R, D, N, T, S** → alphabetically: **B, D, N, R, S, S, T**
New arrangement:
B A D E N R I S O S U T
Compare with the original word, only the **7th letter = I** remains unchanged.
Subject:) CSAT
Topic:) Logical and Verbal Reasoning
Subtopic:)

Q.50) A rectangular display board measures 4.8 m in length and 2.1 m in breadth. Acoustic panels of size 130 cm by 90 cm have to be fixed such that the panels do not overlap. A panel can be placed in any orientation so long as its edges are parallel to the edges of the board. What is the maximum number of panels that can be accommodated on the board?
a) 5
b) 6
c) 7
d) 8

Ans) c
Exp) Option c is the correct answer.
Convert board size to cm:
Board = 480 cm × 210 cm
Panel = 130 cm × 90 cm
A **130 × 90** panel can be stacked only with another **130 × 90** panel, because:
90 + 90 = 180 ≤ 210, but
130 + 90 = 220 > 210 and **130 + 130 > 210**
So across the **480 cm** length, the best packing is:

- **3 columns** of width **130 cm**, each holding **2 panels**
- → **3 × 2 = 6 panels**
- Remaining length = **480 − 3×130 = 90 cm**
- In this **90 cm** strip, we can place **1 panel** of size **90 × 130**

Thus total panels: **6 + 1 = 7**
Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)